DESI DR1 QSO
Direct differential clustering and exploratory BAO inference

Full selected NGC + SGC catalogues · complete random_0 · $0.8 \leq z < 2.1$

A single report of measured correlation functions, first and second redshift derivatives, joint distance posteriors, shape-function diagnostics, contours and numerical checks.

Status of the results

The clustering points are obtained from actual QSO/random pair counts. Distance and shape-function curves are conditional on a finite log-AP redshift expansion, a Kaiser angular model, nuisance priors and a regularized internal covariance. They are not official DESI likelihood results or model-free point derivatives.

Processed: 856,831 selected QSOs and 12,689,095 selected random points. No object thinning. Pre-reconstruction. Analysis date: 8 September 2026.

All figures in this PDF were generated by this QSO run. No earlier LRG constraints, rescaled forecasts or published BAO-distance points were used as fitting data.

Guide: dataset and estimator → measured multipoles → distance likelihood → posterior bands and contours → shape-ratio diagnostics → model and numerical checks.

1. Dataset and directly measured observables

Region	Selected QSOs	Selected randoms	Jackknife regions
NGC	555,913	8,134,597	80
SGC	300,918	4,554,498	40

The inputs are the public DR1 LSS iron v1.5 QSO clustering-ready FITS catalogues and random realization 0 [1]. Standard catalogue correction weights are multiplied by WEIGHT_FKP. The selection is applied identically to data and randoms. NGC and SGC pair counts and normalizations are formed separately before combination.

Accepted pairs	NGC	SGC
DD	95,536,041	44,122,226
DR	2,707,776,169	1,313,692,233
RR	19,328,754,516	9,816,907,597

Pair separations: 40–160 h^{-1} Mpc; bins: 4 h^{-1} Mpc and 20 bins in $|\mu|$. Angular pairs below 0.05 degrees are excluded. Each spatial deletion removes every pair touching its region and recomputes catalogue normalizations and RR moments. The cap-wise jackknife covariance contributions are added; no independent-mock covariance correction is asserted.

Raw joint covariance rank: 118 for 270 entries. This matrix is not inverted at its full dimension.

2. Derivative estimators and resolution

In each separation/angle cell, let u be pair midpoint redshift minus its RR-weighted mean and let $m_n = \langle u^n \rangle RR$. The baseline random-pair density K integrates to one. All numerator pair counts use the same catalogue normalizations; signed moments are never independently normalized.

$$q_1 = u / m_2$$

$$q_2 = 2 [u^2 - m_2 - (m_3/m_2)u] / [m_4 - m_2^2 - m_3^2/m_2]$$

Measured RR moments through sixth order supply the second-derivative normalization and resolution moments. The q_2 weight has zero constant and linear response, including the skewness correction. Pair-count and analytic polynomial-response tests are included with the reproduction products [2].

$$\xi[q_1] = \int W_1(z)\, \partial\xi/\partial z\, dz; \quad \xi[q_2] = \int W_2(z)\, \partial^2\xi/\partial z^2\, dz$$

These integration-by-parts identities do not require a Taylor model for $\xi(z)$. Their kernels vary with separation and angle; they should not be interpreted as delta functions at an effective redshift. The plots use s^2 times the corresponding measured multipole.

Kernel diagnostic	First derivative	Second derivative
RR-weighted mean redshift	1.46407	1.45603
Square root of mean kernel variance	0.26827	0.23047

The second-derivative statistic measures curvature of the full clustering signal: geometrical evolution, bias, growth, redshift-space distortions, damping and residual selection effects all contribute. It is not by itself D_H″ or D_M″.

3. Joint BAO model and uncertainty conventions

The reference oscillatory spectrum is calculated using CAMB and separated from a smooth spectrum. The BAO part is AP-remapped with independent transverse and radial distances, anisotropically damped, transformed to configuration space and averaged through measured RR redshift kernels and separation bins. The smooth baseline has independent broadband freedom. No published QSO BAO distances enter the fit [1,3].

$$u = (z - 1.49) / 0.65; \ln \alpha_A(z) = a_{A,0} + a_{A,1} u + \tfrac{1}{2} a_{A,2} u^2$$

The transverse and radial coefficient sets are independent. Both distance normalizations are free. The primary quadratic fit uses ξ, ξ' and ξ''; the comparison linear fit omits the quadratic AP terms and uses ξ and ξ'. This comparison changes both the model and the channels, so it is not an information-gain measurement.

The amplitude is positive, $A(z)$, with Kaiser angular form $A(z)[1+\beta(z)\mu^2]^2$. Log amplitude has quadratic redshift freedom; β is linear and remains positive. Effective damping priors are $\Sigma_\perp = 3 \pm 1$ and $\Sigma_\parallel = 8 \pm 3\ h^{-1}$ Mpc. They approximate, rather than separately calibrate, QSO radial smearing. All hard bounds and broadband priors are listed in the machine-readable summaries.

Fitting uses 8 h^{-1} Mpc bins over 52–148 h^{-1} Mpc and the monopole/quadrupole. $C_{used} = 0.8\ C_{JK} + 0.2\ \mathrm{diag}(C_{JK})$, with best-fit sensitivity checks at diagonal shrinkage 0.1 and 0.4. Figure error bars are raw 1σ jackknife diagonal errors. Shaded distance/shape bands are pointwise posterior quantiles, not simultaneous confidence envelopes.

Two independently seeded ensembles are run for each redshift model. The report gives autocorrelation and split-chain diagnostics but does not claim mock-calibrated coverage or a discovery significance. Redshift quadrature is checked against the original 0.01-spaced RR histogram.

4. Distance and differential-distance results

All values below are dimensionless BAO distances or their derivatives per unit redshift, evaluated from the fitted functions at $z_p = 1.49$. Brackets contain central 68% marginal posterior intervals. The two fits use the same catalogue; they are not independent measurements.

Quantity	Linear model + ξ, ξ′	Quadratic model + ξ, ξ′, ξ″
D_M / r_d	30.654 [29.497, 31.827]	31.488 [30.184, 32.815]
D_H / r_d	12.118 [11.384, 12.878]	12.066 [11.244, 12.990]
$d(D_M/r_d)/dz$	17.676 [13.754, 21.958]	16.888 [12.612, 20.312]
$d(D_H/r_d)/dz$	-7.612 [-10.573, -4.370]	-6.528 [-9.263, -3.969]
$d^2(D_M/r_d)/dz^2$	-2.034 [-6.394, 3.731]	-21.839 [-43.564, 2.088]
$d^2(D_H/r_d)/dz^2$	6.423 [3.202, 10.820]	11.474 [-1.443, 21.252]

The second derivatives in the linear-AP fit follow from that restricted model; they are not independent measurements of the second-derivative channel. Neither fit imposes $X' = Y$, where $X = D_M/r_d$ and $Y = D_H/r_d$.

Derived quantity	Linear	Quadratic
DV_over_rd	25.697 [25.062, 26.337]	26.173 [25.435, 26.847]
q_pivot	0.569 [-0.095, 1.175]	0.344 [-0.172, 0.873]
dDMdz_minus_DH	5.522 [1.565, 9.847]	4.656 [0.426, 8.257]

The published DR1 QSO result is isotropic $D_V/r_d = 26.07 \pm 0.67$ at $z_{eff} = 1.49$ [1]. It does not provide official QSO dD_M/dz or dD_H/dz measurements. Our context comparison differs in reconstruction, catalogue version, model and covariance, and shares galaxies with the official measurement; no independent-data discrepancy p-value is calculated.

5. Shape functions: diagnostic, not an unqualified detection

Use $F(a) = a^3/[D_H(a)/r_d]^2$ and $x = \ln a$. To avoid extrapolating QSOs to the unobserved present-day boundary, normalize S_0 and S_1 at $a_p = 1/(1+1.49)$, not $a = 1$ [4].

$$S_{0,p} = (a/a_p)^3 - 3[F(a) - F(a_p)] / F_x(a_p)$$

$$S_{1,p} = (a_p/a)^3 F_x(a) / F_x(a_p); \quad S_2 = -F_{xx}(a) / [3F_x(a)]$$

The flat matter+Λ reference has $S_{0,p} = S_{1,p} = 1$ and $S_2 = -1$. $S_{0,p}$ and $S_{1,p}$ have exactly zero spread at the pivot by definition: that pinch is not a precision measurement. The paper's present-day-normalized S_0 and S_1 cannot be obtained from this QSO-only interval without a continuation assumption or lower-redshift data.

Each posterior sample is transformed analytically; F_x is not forced to remain positive. Denominator zero crossings can generate poles and heavy tails. The figures show signed-log axes and posterior quantiles together with geometrical-prior quantiles. No arbitrary cut removes singular histories. Means and Gaussian variances are not asserted for these ratio distributions.

Pole/sensitivity diagnostic	Linear	Quadratic
Posterior fraction with F_x sign crossing	31.7%	79.7%
Prior fraction with F_x sign crossing	21.0%	71.5%

A visually structured median of a broad or singular ratio distribution is not evidence for dynamical dark energy. Distances and derivatives remain conditional on the explicit log-AP basis and nuisance model; the shape plots expose, rather than hide, that limitation.

6. Numerical validation and limitations

Diagnostic	Linear	Quadratic
Estimated effective sample count	1393	1290
Largest independent-ensemble median shift / posterior σ	0.088	0.053
Largest redshift-quadrature change / raw data σ	0.0491	0.0253
Maximum parameter boundary fraction	5.23%	4.45%

The exact counter is verified against exhaustive deterministic pair enumeration, including all delete-one contributions, all seven RR moments and disjoint partition completeness. The analytic polynomial-response test checks the unit normalization of ξ'' using the measured RR moments. These are algorithm tests, not cosmological mock validation.

Remaining limitations: pre-reconstruction; one complete random realization rather than a suite; internal jackknife covariance and shrinkage; finite redshift and angular discretization; Gaussian effective radial damping; nuisance-model dependence; and no injected-cosmology mock coverage tests. Same-galaxy cross-covariances are required before combining this compression with official DESI constraints.

Robust deliverable: the actual full-sample ξ, ξ' and normalized ξ'' measurements, their shared covariance and RR response information. Exploratory deliverable: joint BAO and shape-function posterior diagnostics with transparent priors and conditioning.

7. Sources and reproducibility

[1] DESI DR1 catalogues and BAO analysis
https://data.desi.lbl.gov/public/dr1/survey/catalogs/dr1/LSS/iron/LSScats/v1.5/

[1a] DESI 2024 III: BAO from galaxies and quasars
https://arxiv.org/abs/2404.03000

[2] Optimal redshift weighting and validation
https://arxiv.org/abs/1411.1424

[2a] Redshift-weighted BAO after reconstruction
https://arxiv.org/abs/1604.01050

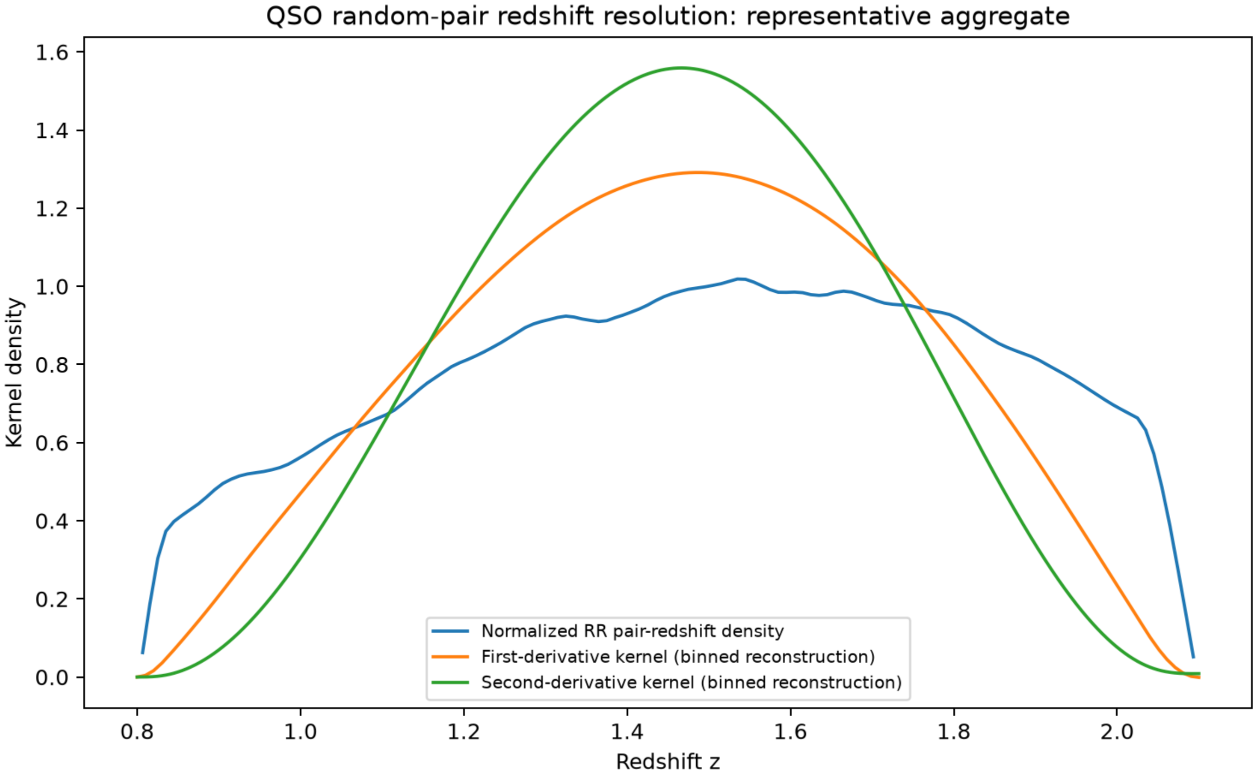
[3] CAMB
https://camb.readthedocs.io/

[4] Shape-function definitions, Eq. (5)
https://arxiv.org/abs/2504.06118

Reproduction files are in the results archive: adapted pair counter, reducer, QSO likelihood, MCMC driver and this report builder. Input file SHA-256 checksums and selected-row counts are retained in measurement_summary.json; model summaries contain all bounds, priors, numerical diagnostics and covariance sensitivity checks.

The source branch is chatgpt-qso-dr1-analysis-20260908 in icosmology/BAO-DE-fit. Main was not changed. Figures are generated from this run’s saved arrays and posteriors. The archived input catalogues are public DESI products; any scientific publication must include DESI’s required release citations and acknowledgments.

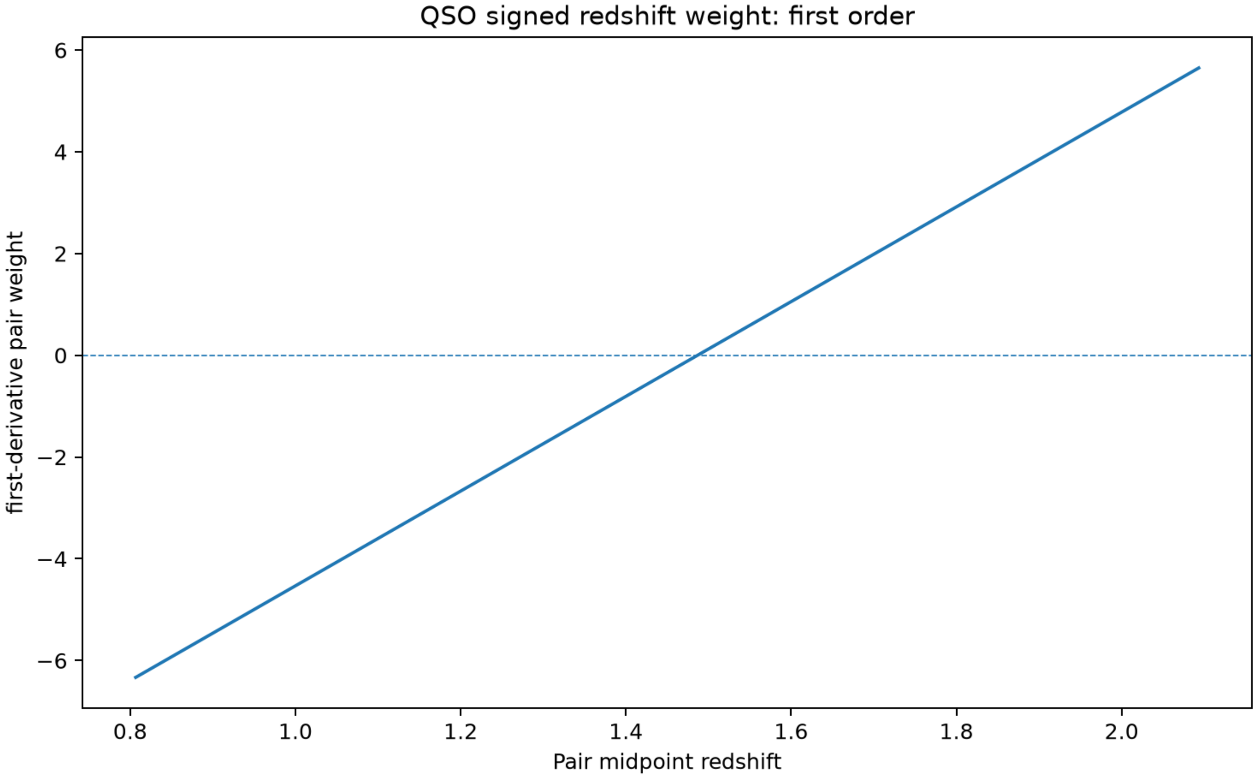
Figure 1 | redshift kernels



Representative RR-weighted aggregate for display. The actual estimator and likelihood retain separation- and angle-dependent redshift responses. Exact normalization uses measured moments, not the plotted binned approximation.

redshift_kernels.png

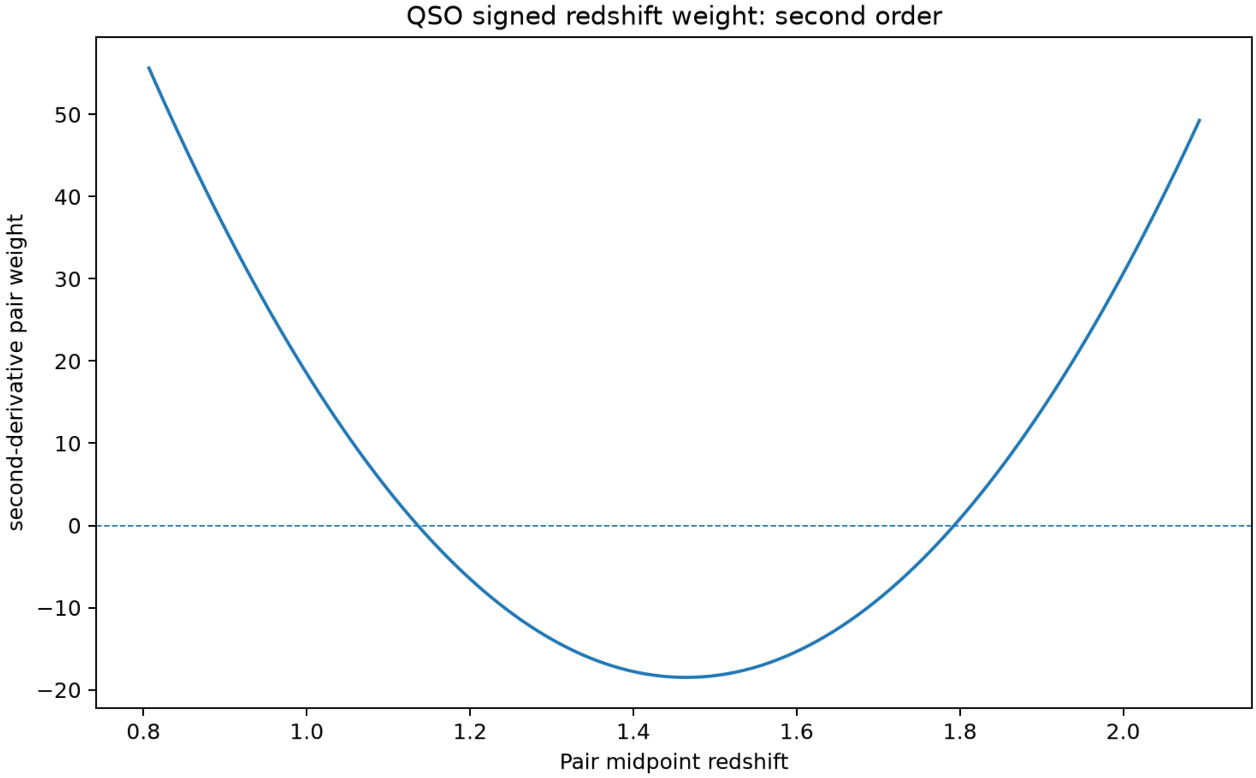
Figure 2 | weight first



Representative RR-weighted aggregate for display. The actual estimator and likelihood retain separation- and angle-dependent redshift responses. Exact normalization uses measured moments, not the plotted binned approximation.

weight_first.png

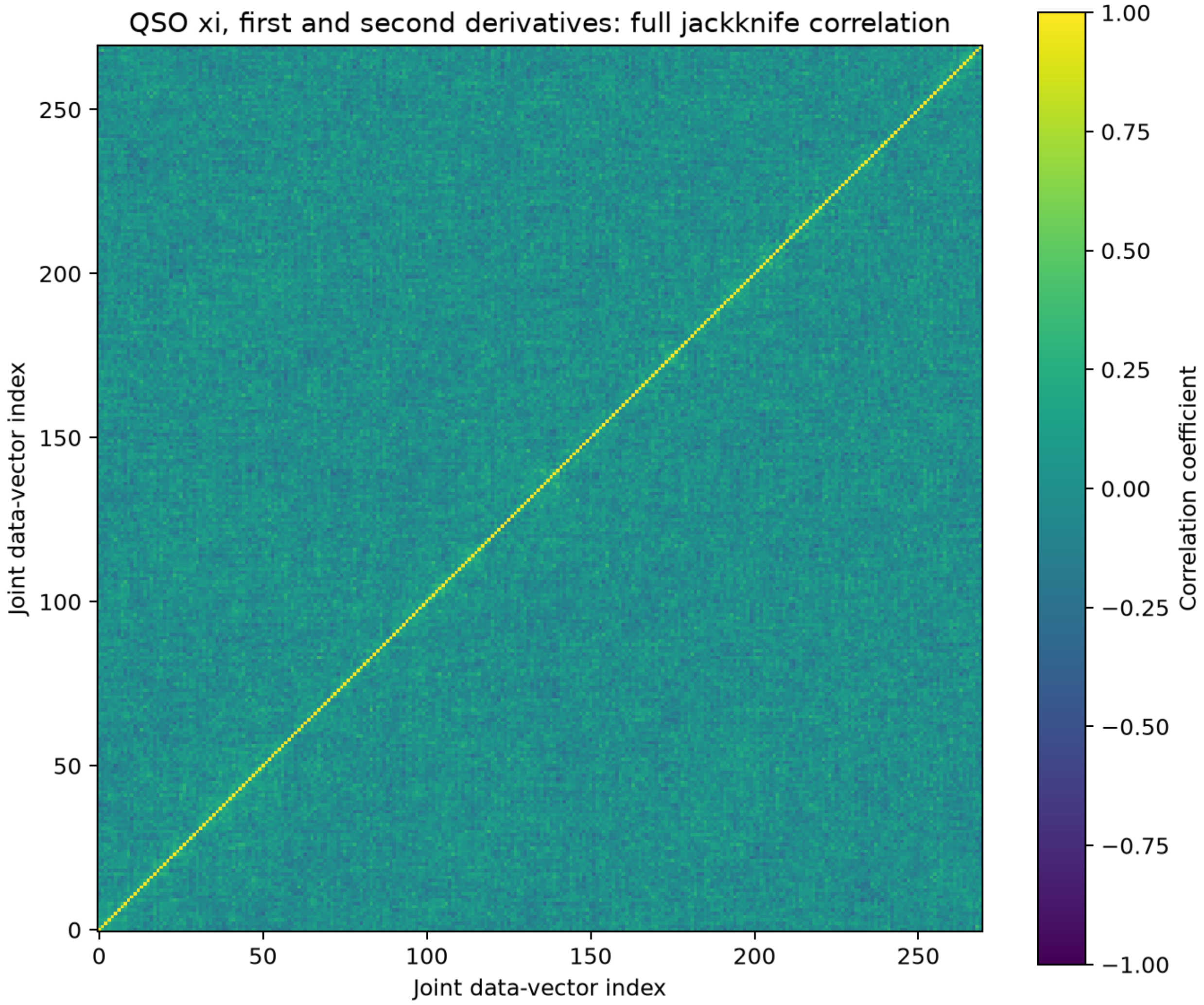
Figure 3 | weight second



Representative RR-weighted aggregate for display. The actual estimator and likelihood retain separation- and angle-dependent redshift responses. Exact normalization uses measured moments, not the plotted binned approximation.

weight_second.png

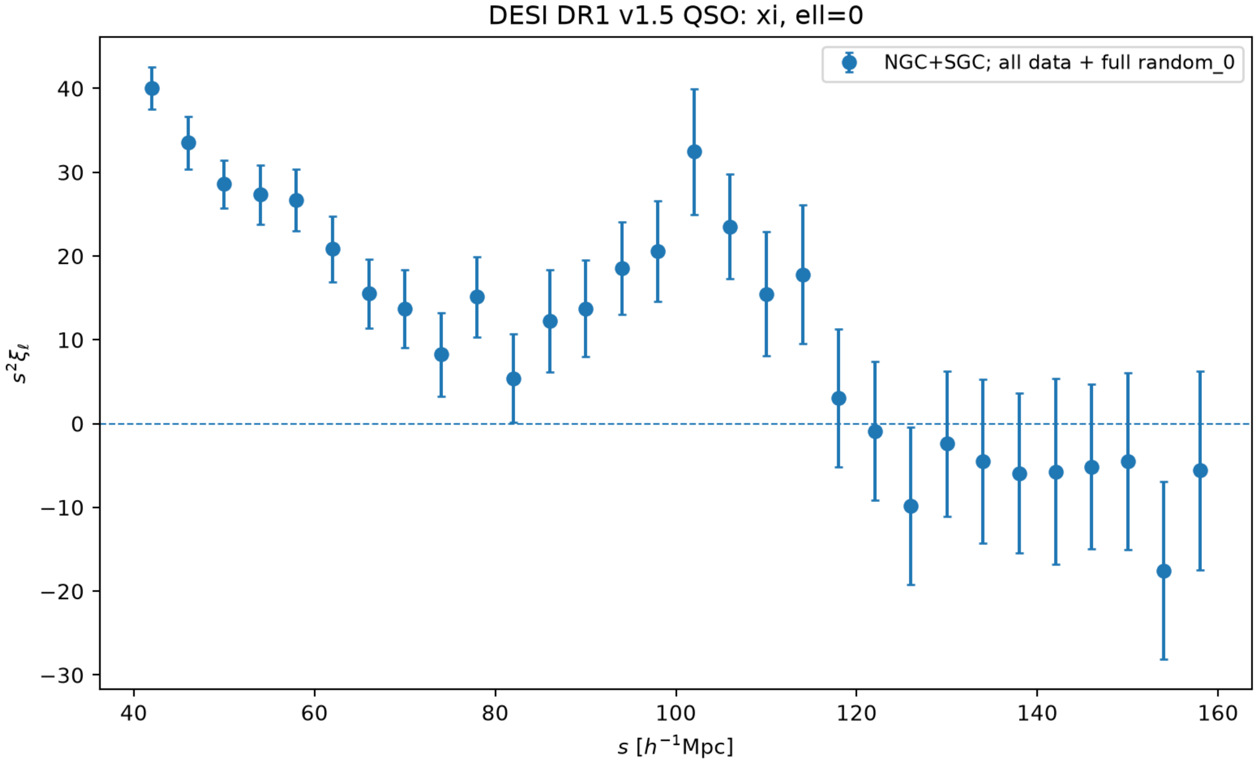
Figure 4 | joint correlation matrix



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

joint_correlation_matrix.png

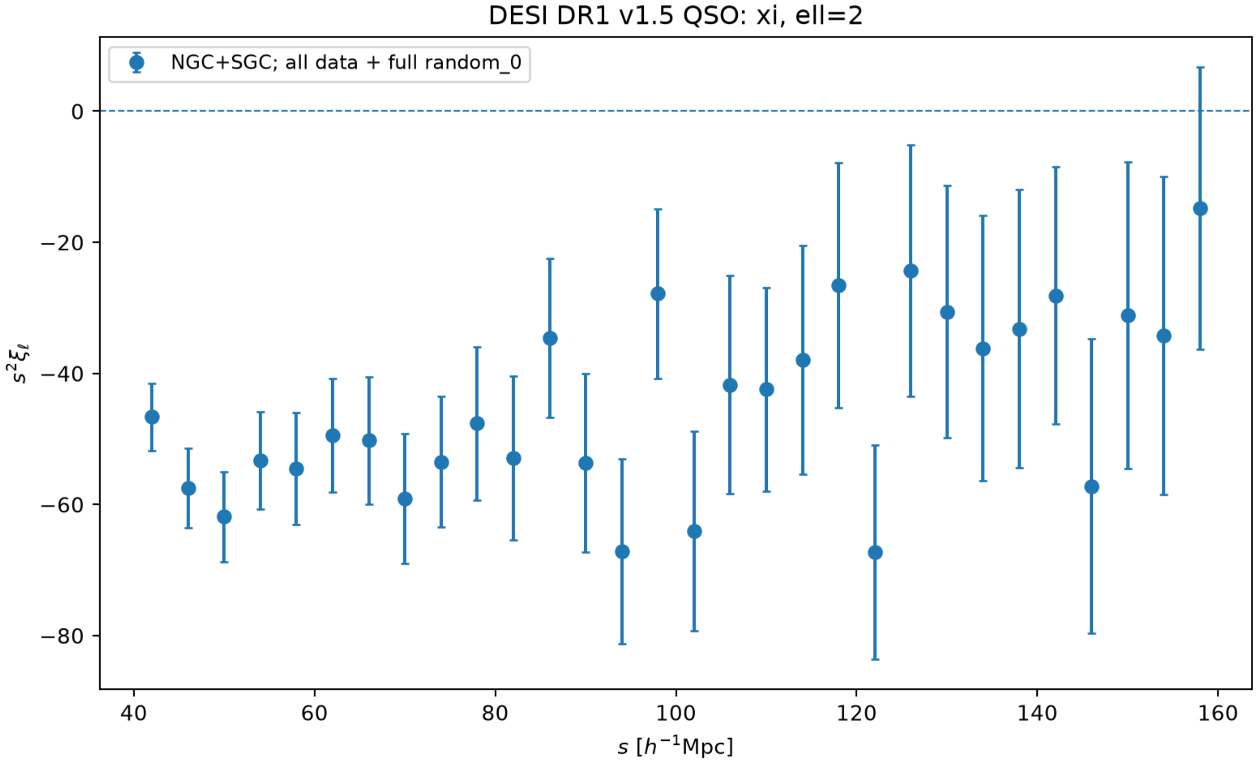
Figure 5 | xi ell0



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

xi_ell0.png

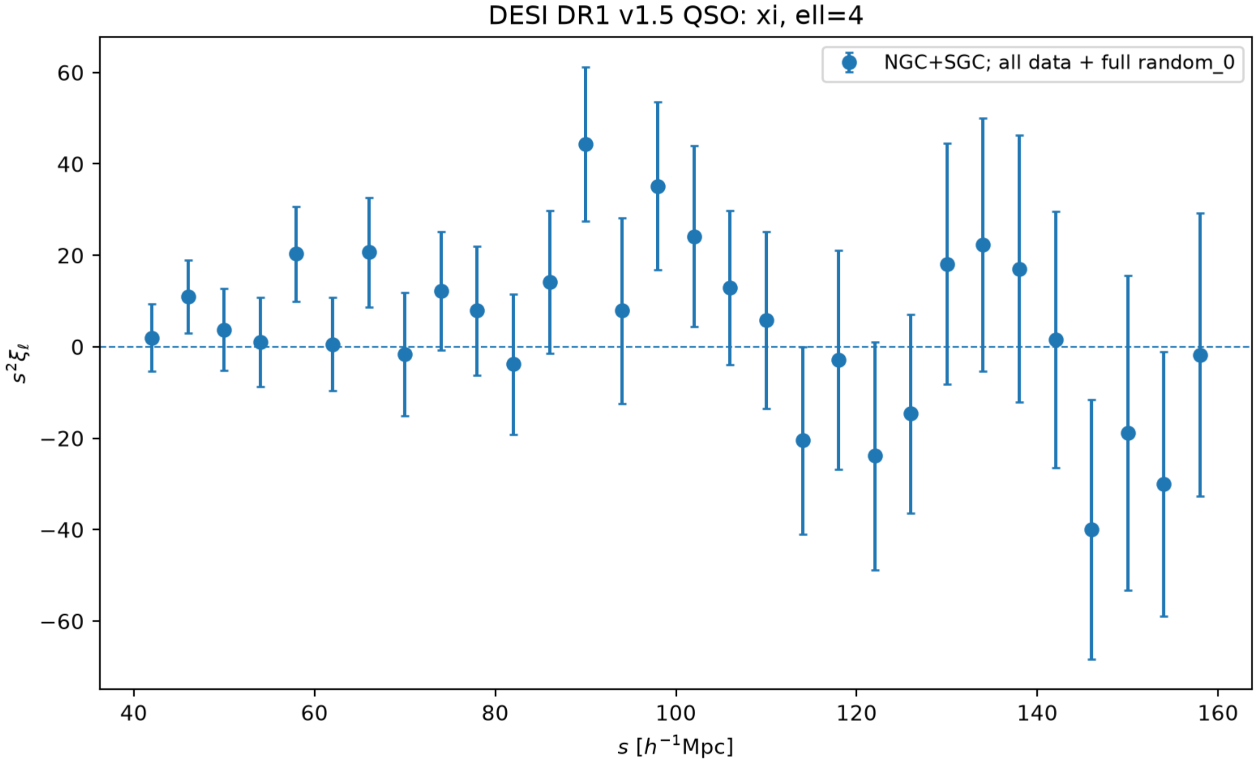
Figure 6 | xi ell2



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

xi_ell2.png

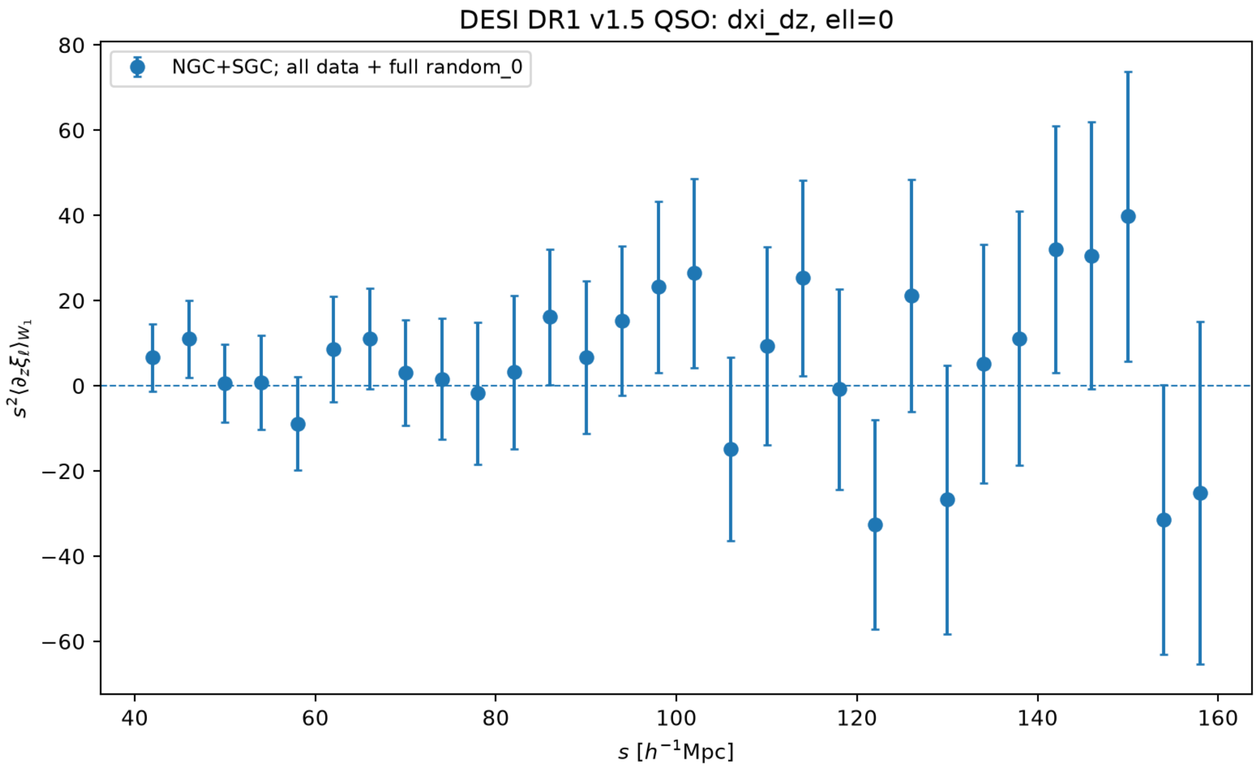
Figure 7 | xi ell4



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

xi_ell4.png

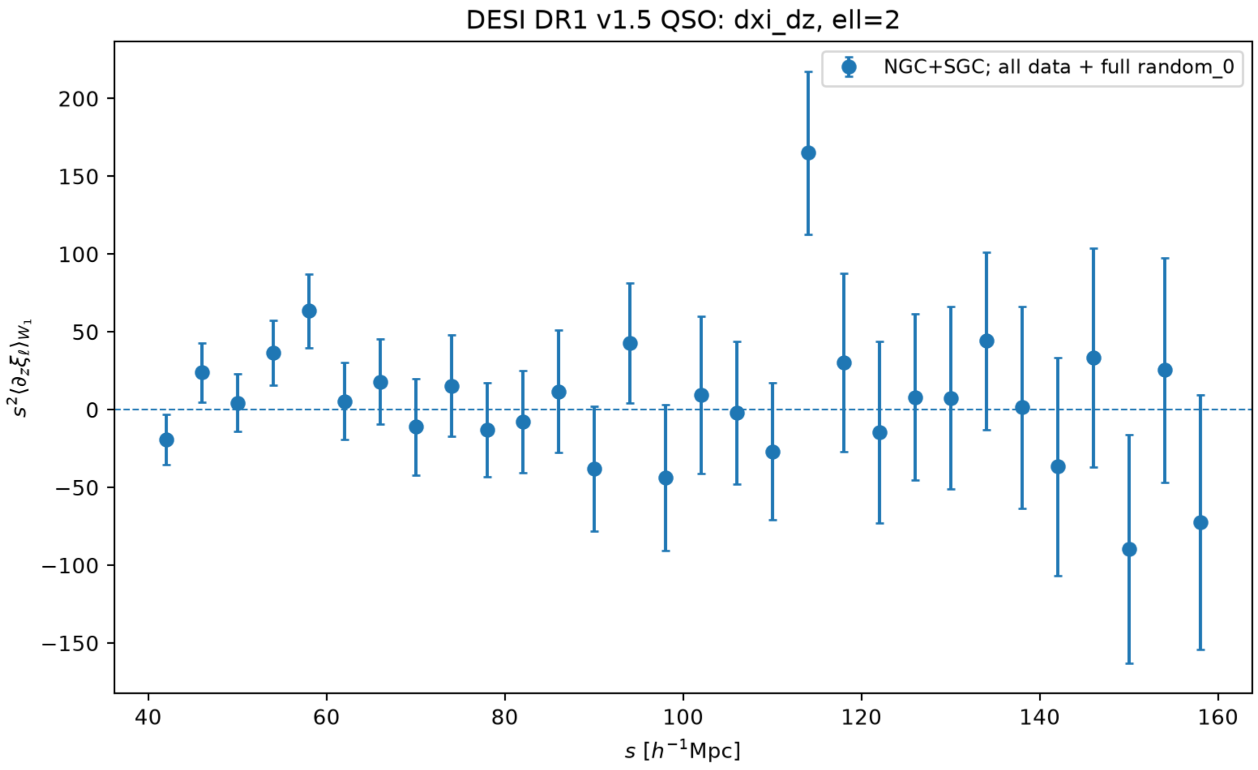
Figure 8 | dxi dz ell0



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

dxi_dz_ell0.png

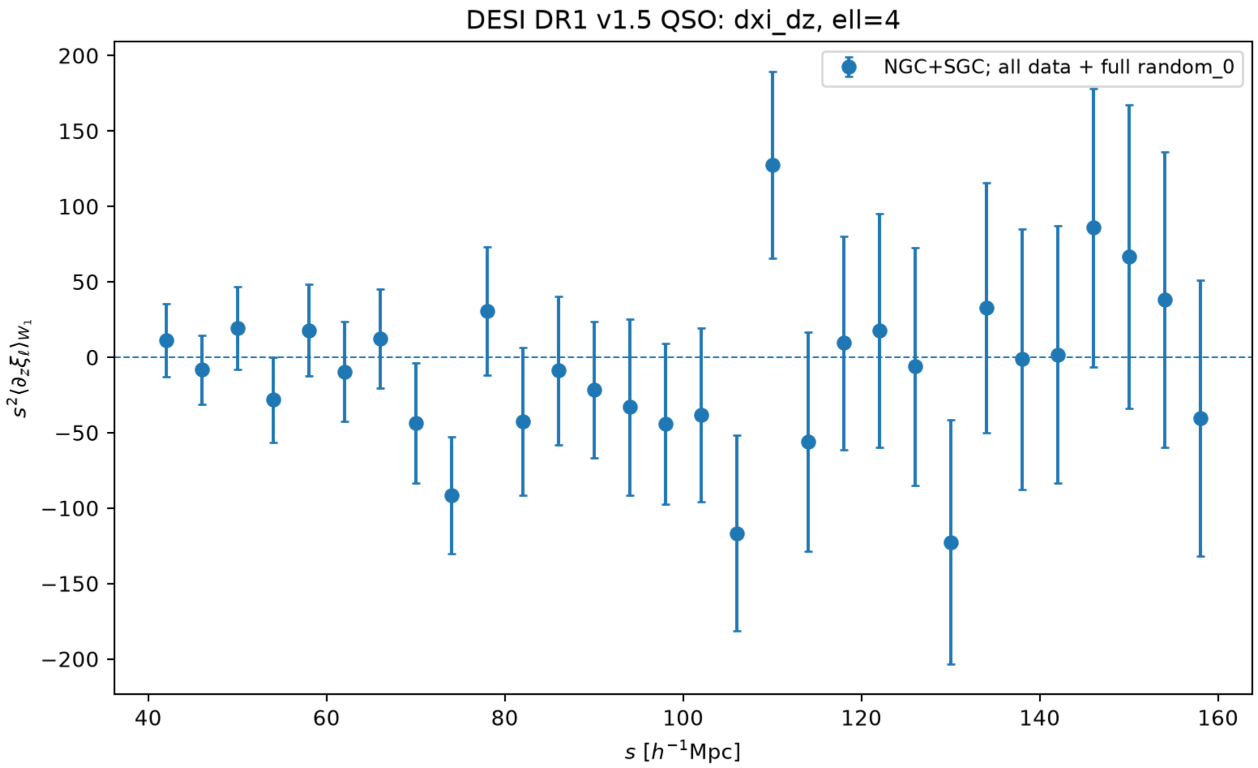
Figure 9 | dxi dz ell2



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

dxi_dz_ell2.png

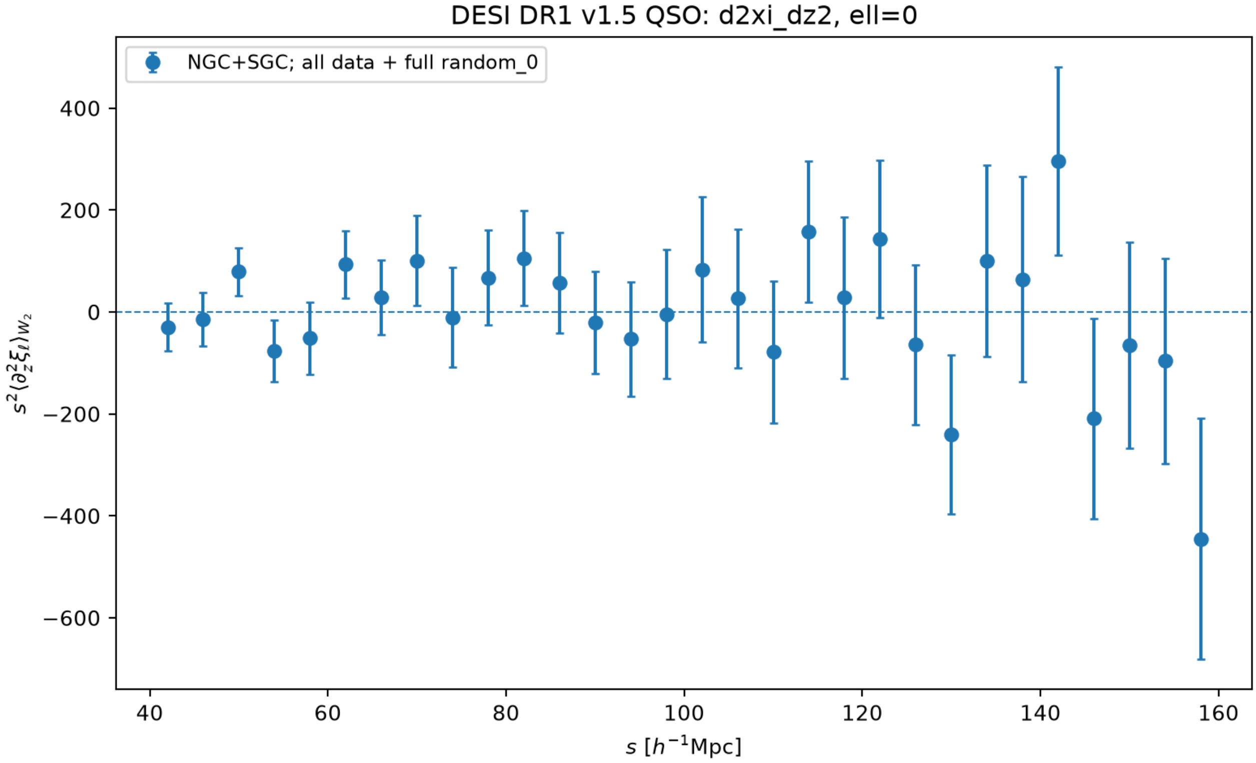
Figure 10 | dxi dz ell4



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

dxi_dz_ell4.png

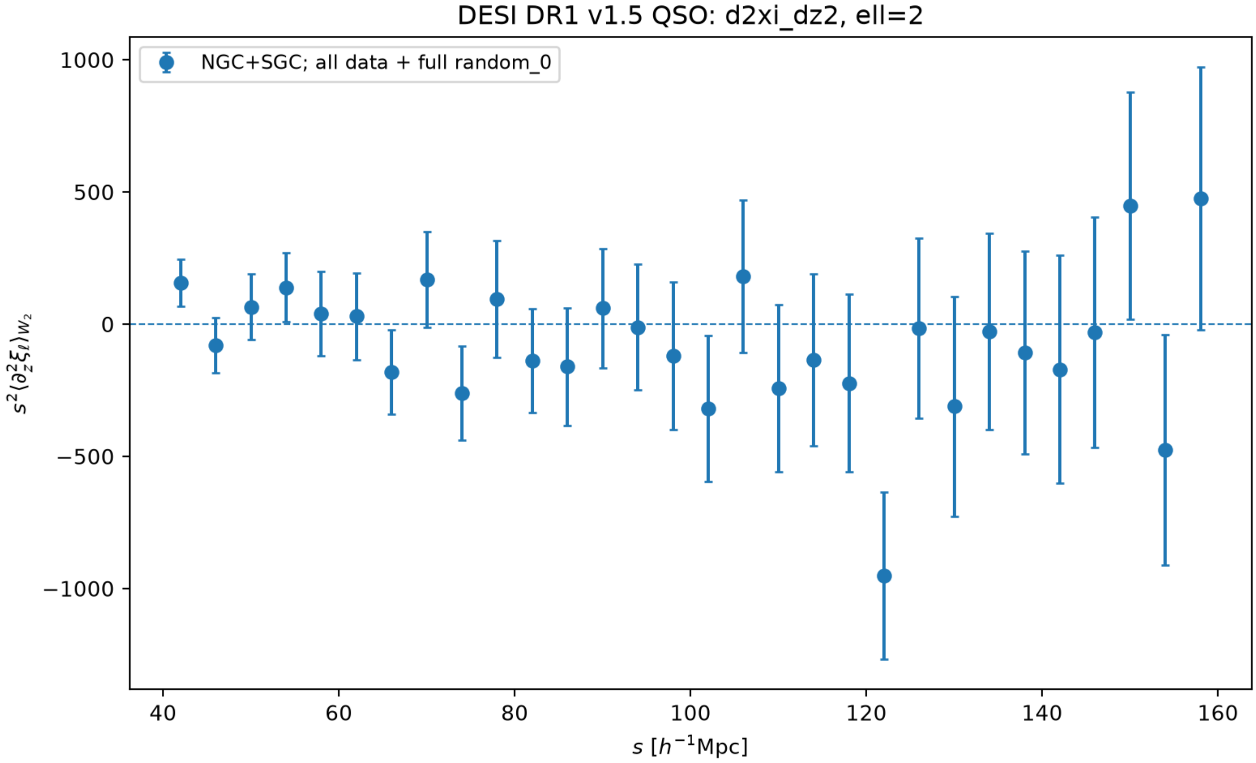
Figure 11 | d2xi dz2 ell0



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

d2xi_dz2_ell0.png

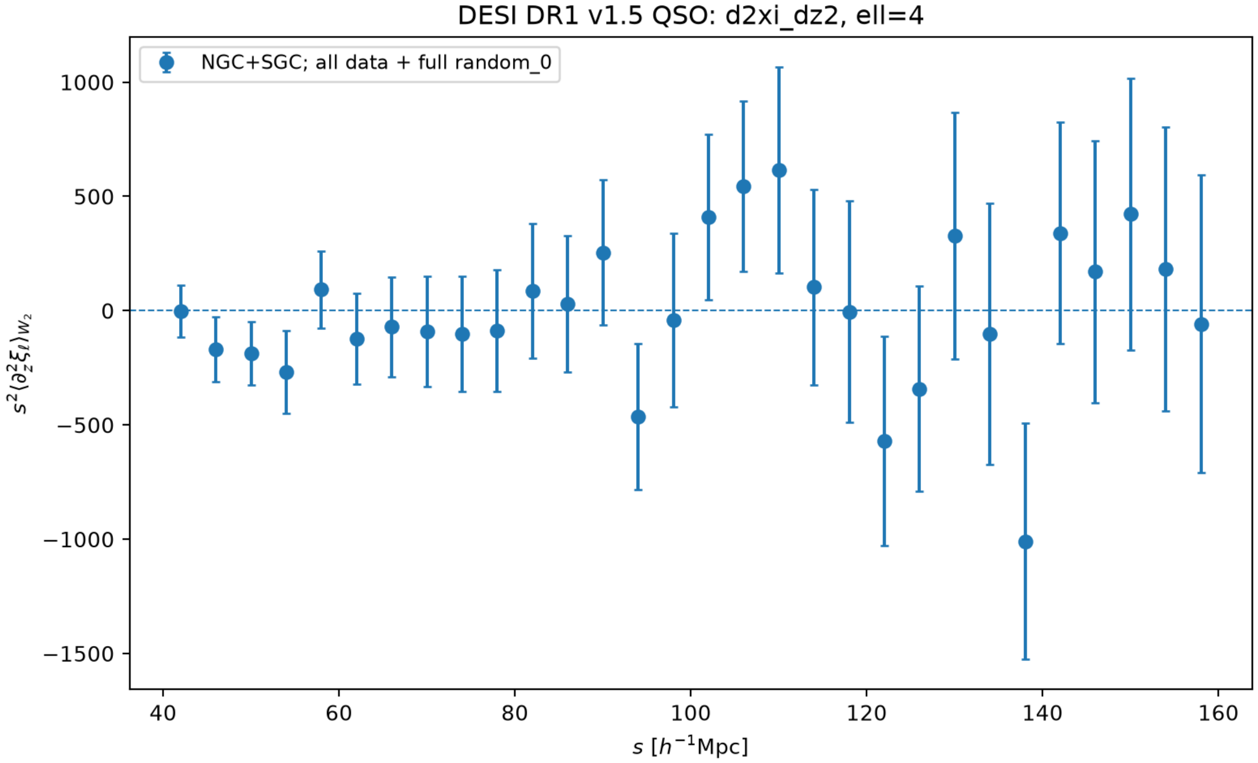
Figure 12 | d2xi dz2 ell2



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

d2xi_dz2_ell2.png

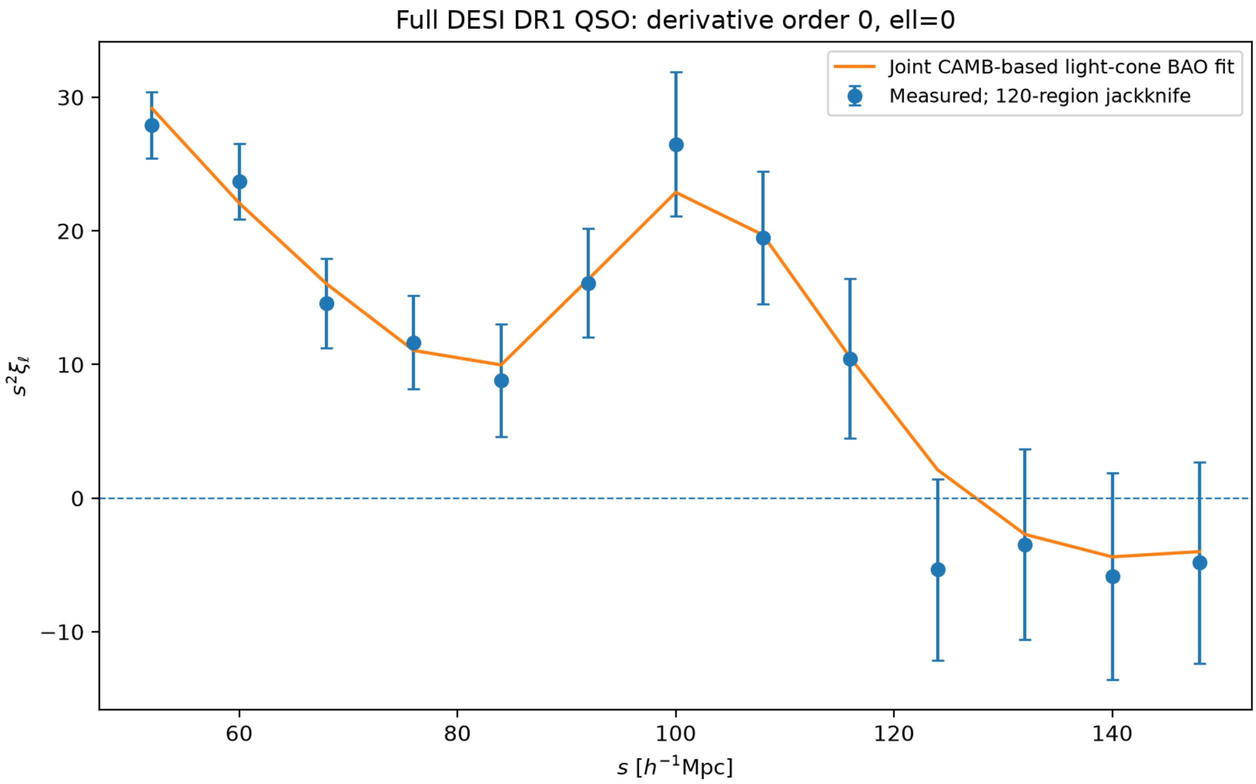
Figure 13 | d2xi dz2 ell4



Actual full-sample QSO pair-count measurement. Error bars are 1σ diagonal uncertainties from the shared 120-region spatial jackknife covariance. Neighbouring bins and derivative channels are correlated. No signal was simulated.

d2xi_dz2_ell4.png

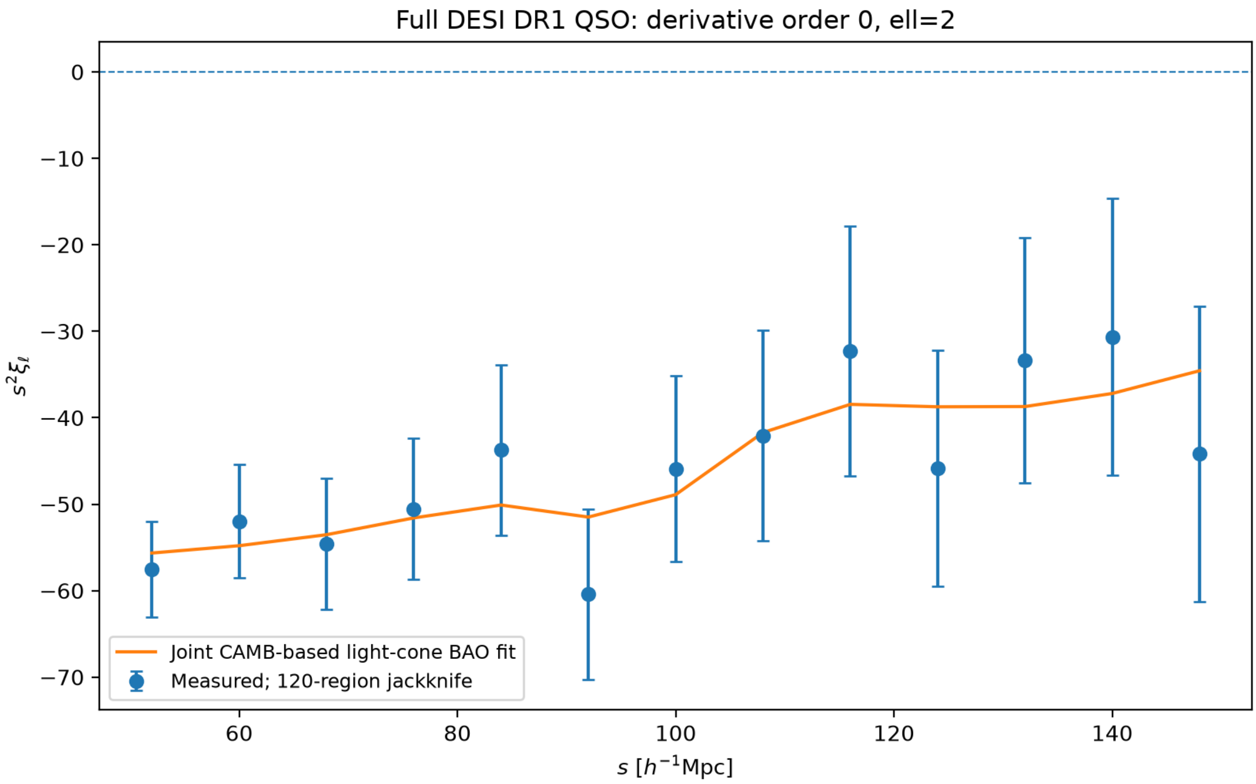
Figure 14 | quadratic fit order0 ell0



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order0_ell0.png

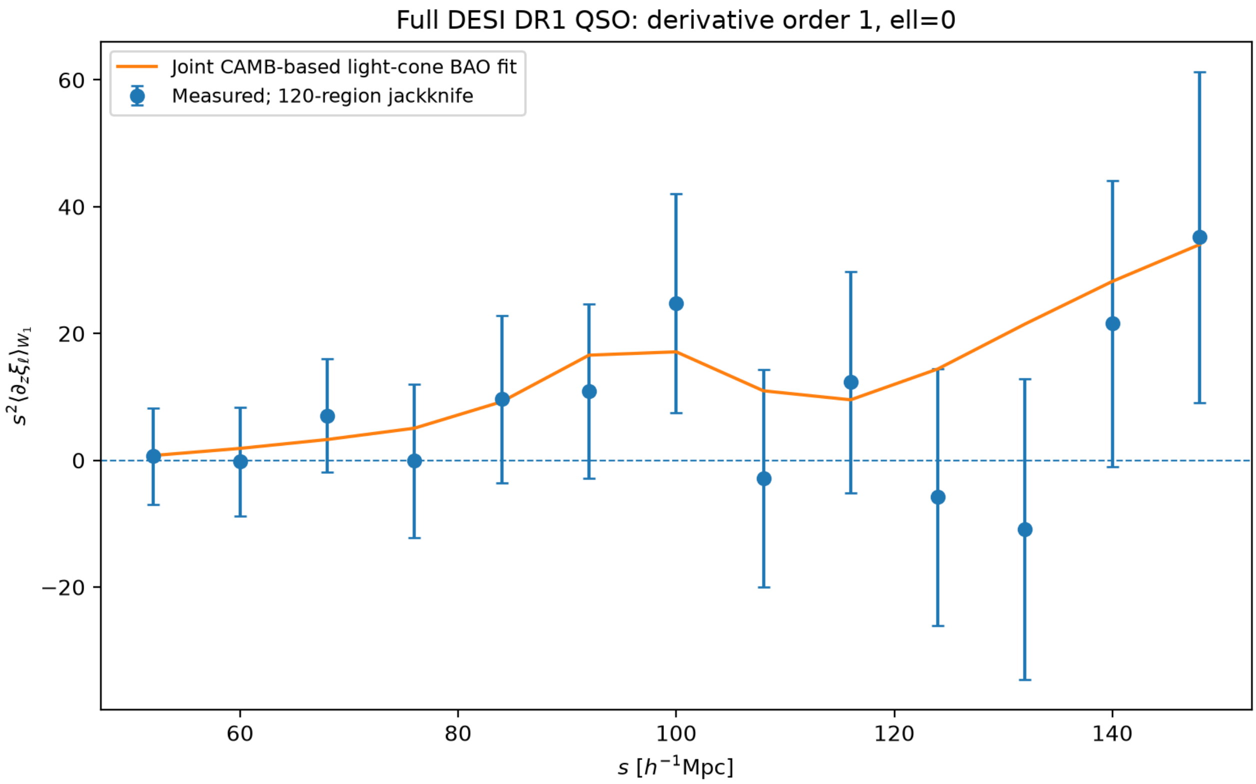
Figure 15 | quadratic fit order0 ell2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order0_ell2.png

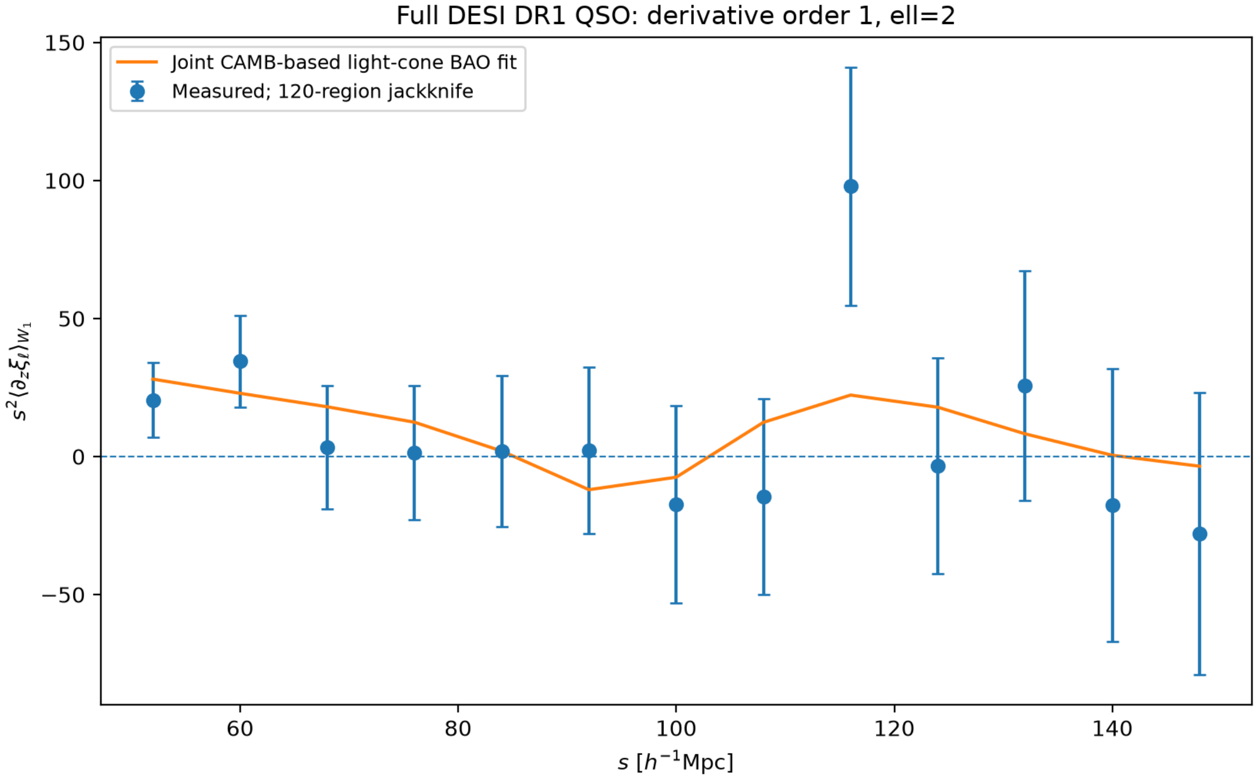
Figure 16 | quadratic fit order1 ell0



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order1_ell0.png

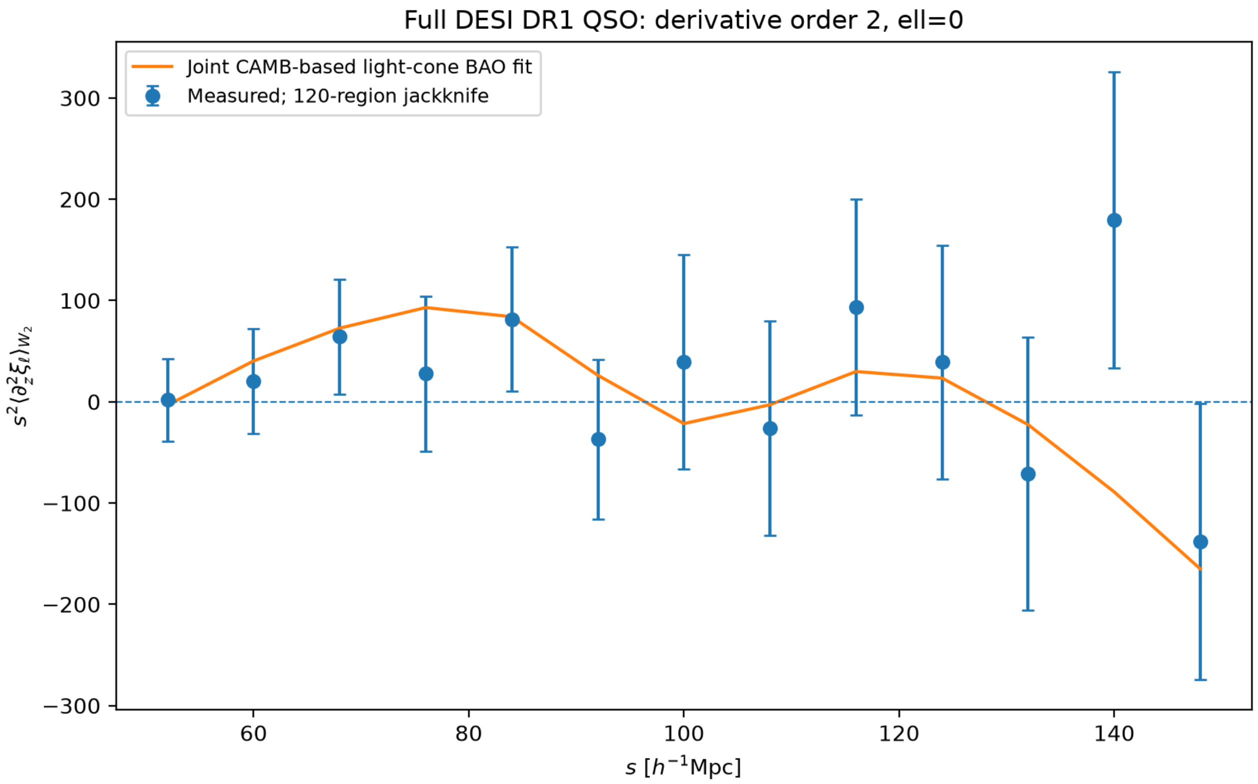
Figure 17 | quadratic fit order1 ell2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order1_ell2.png

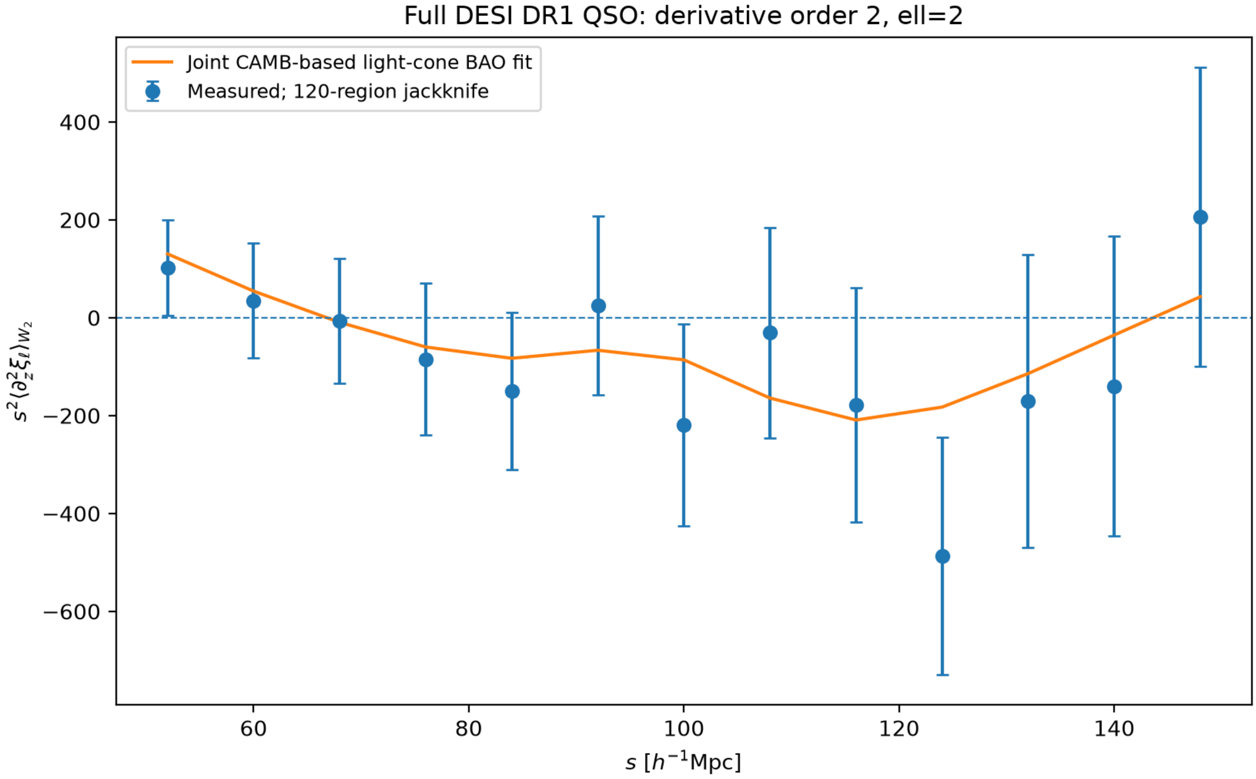
Figure 18 | quadratic fit order2 ell0



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order2_ell0.png

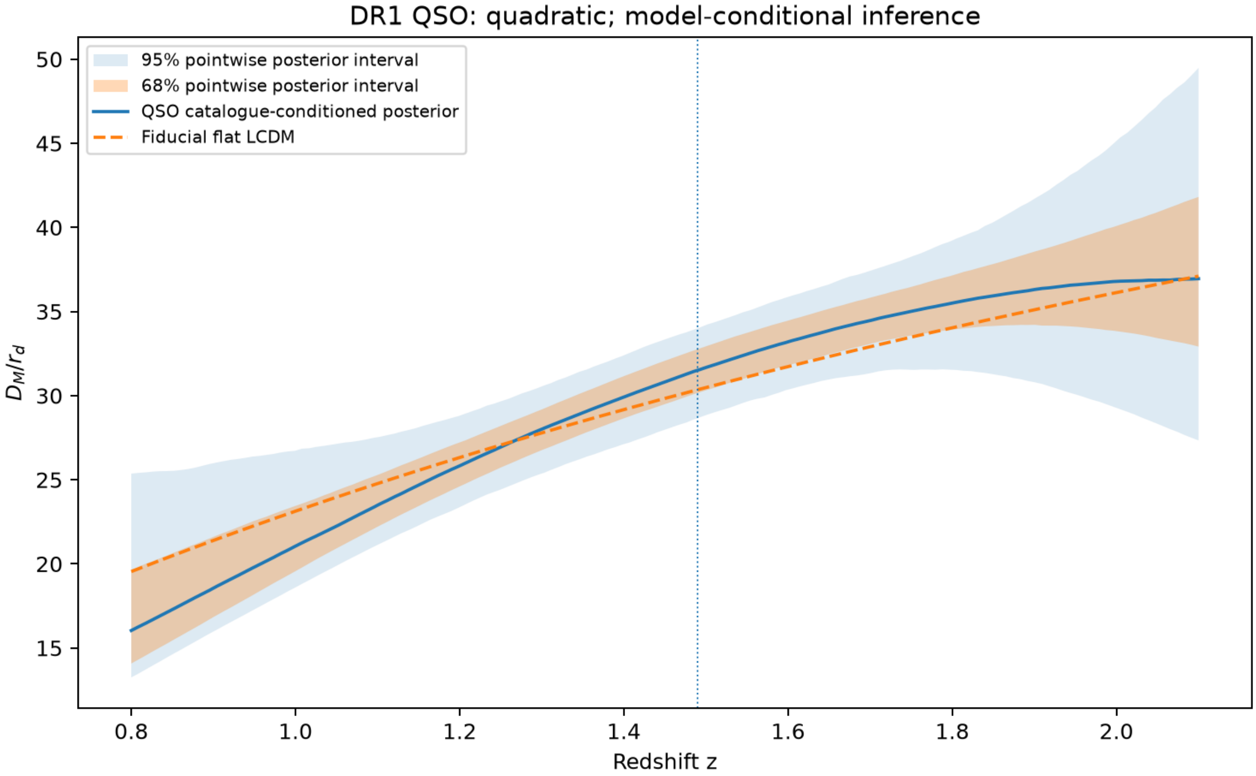
Figure 19 | quadratic fit order2 ell2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_fit_order2_ell2.png

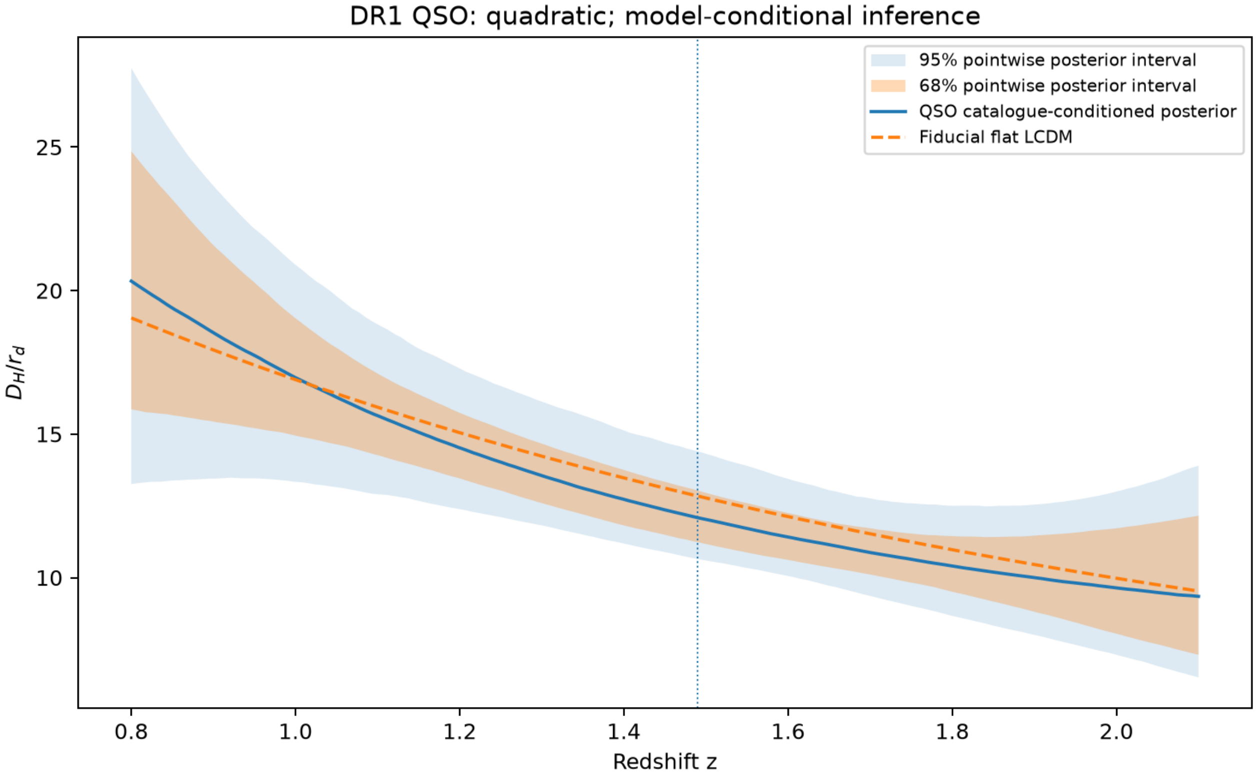
Figure 20 | quadratic DM



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_DM.png

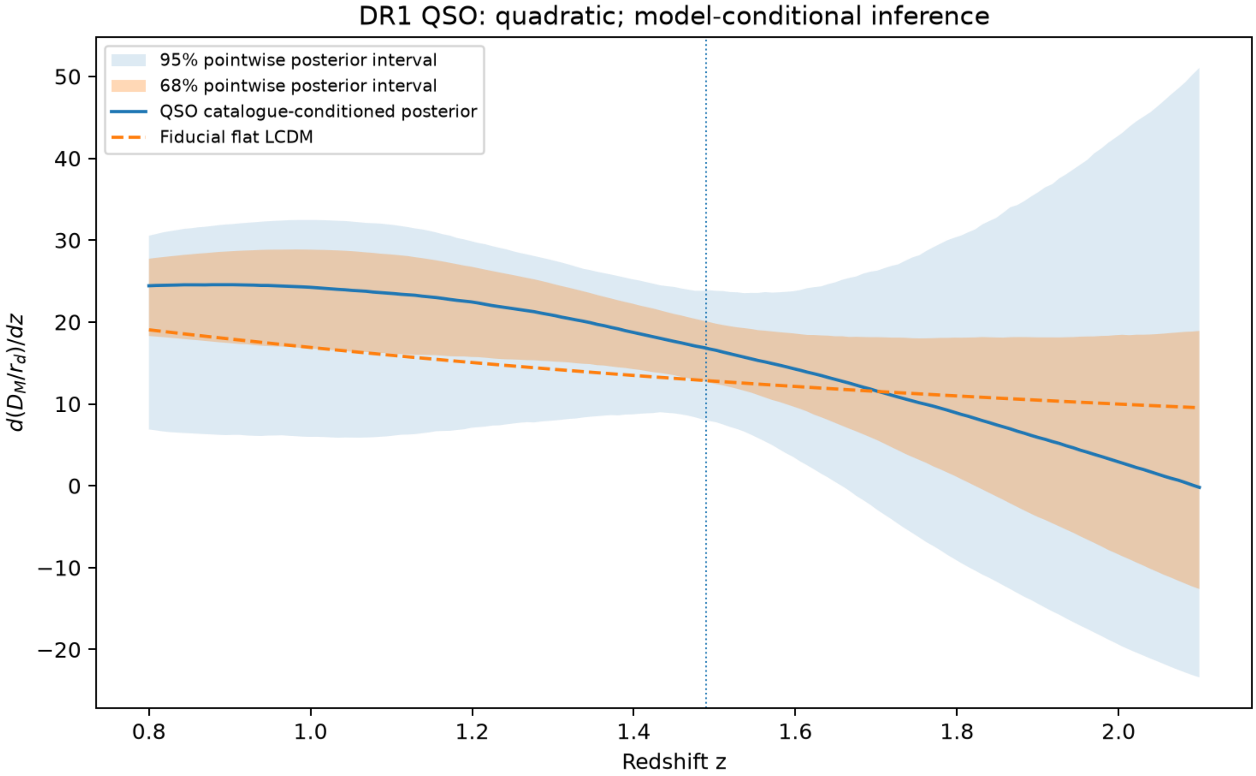
Figure 21 | quadratic DH



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_DH.png

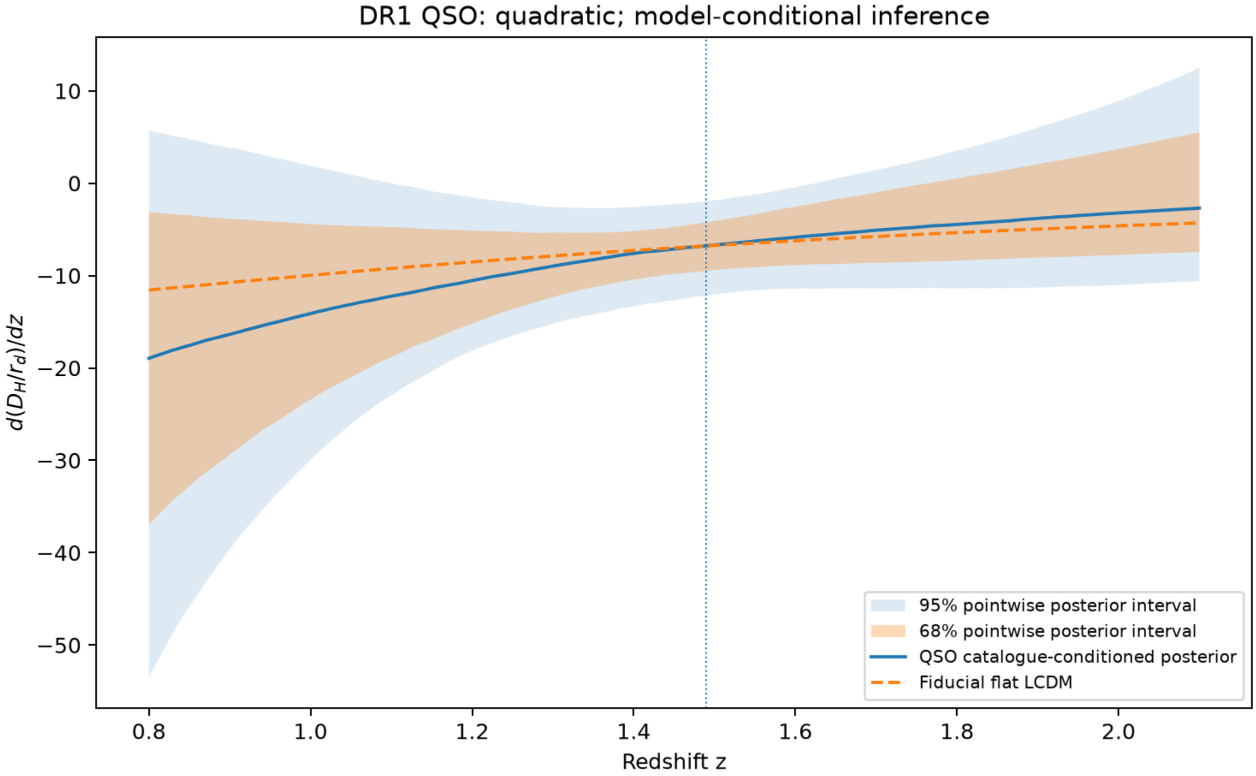
Figure 22 | quadratic dDM dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_dDM_dz.png

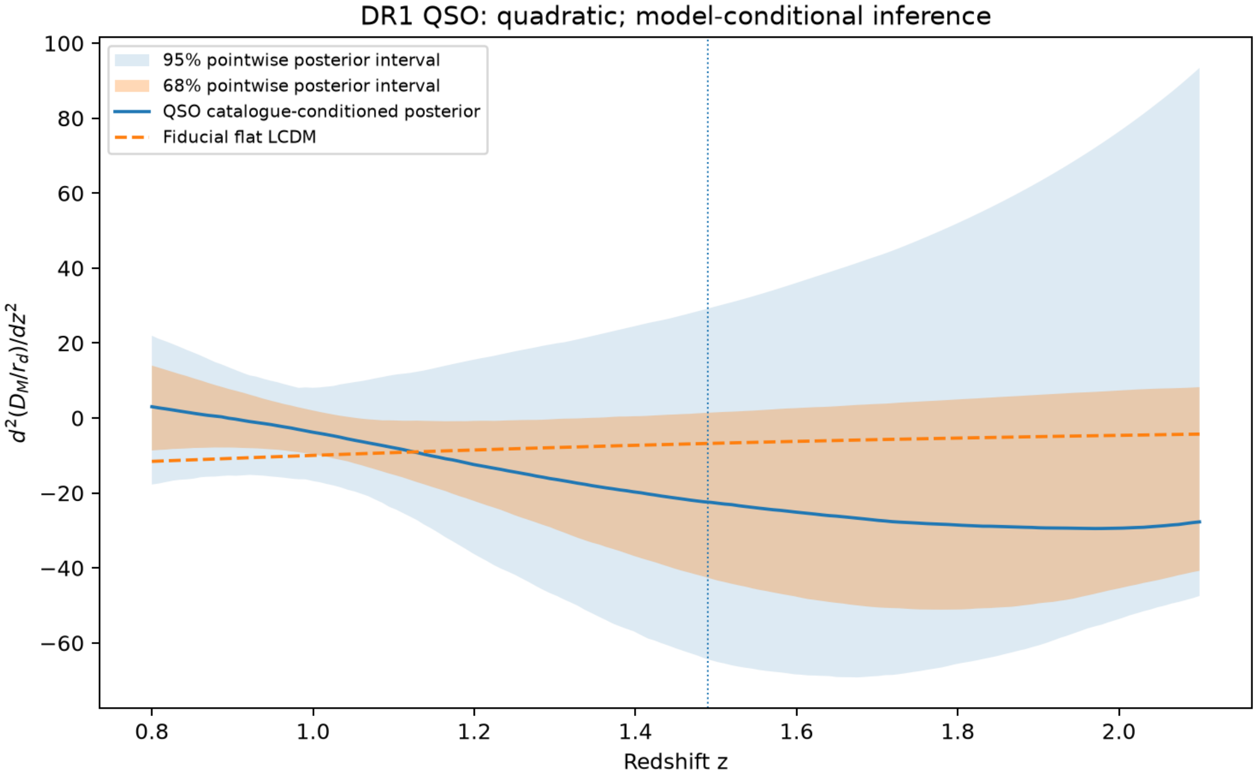
Figure 23 | quadratic dDH dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_dDH_dz.png

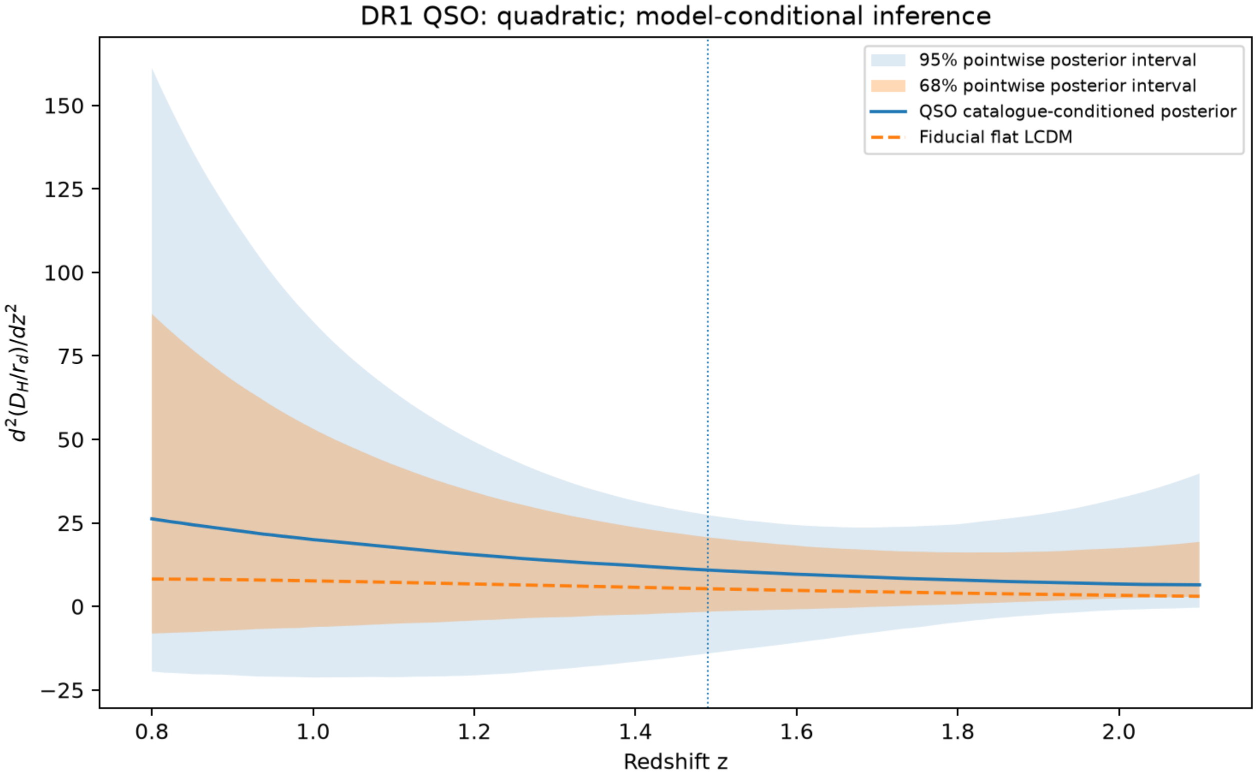
Figure 24 | quadratic d2DM dz2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_d2DM_dz2.png

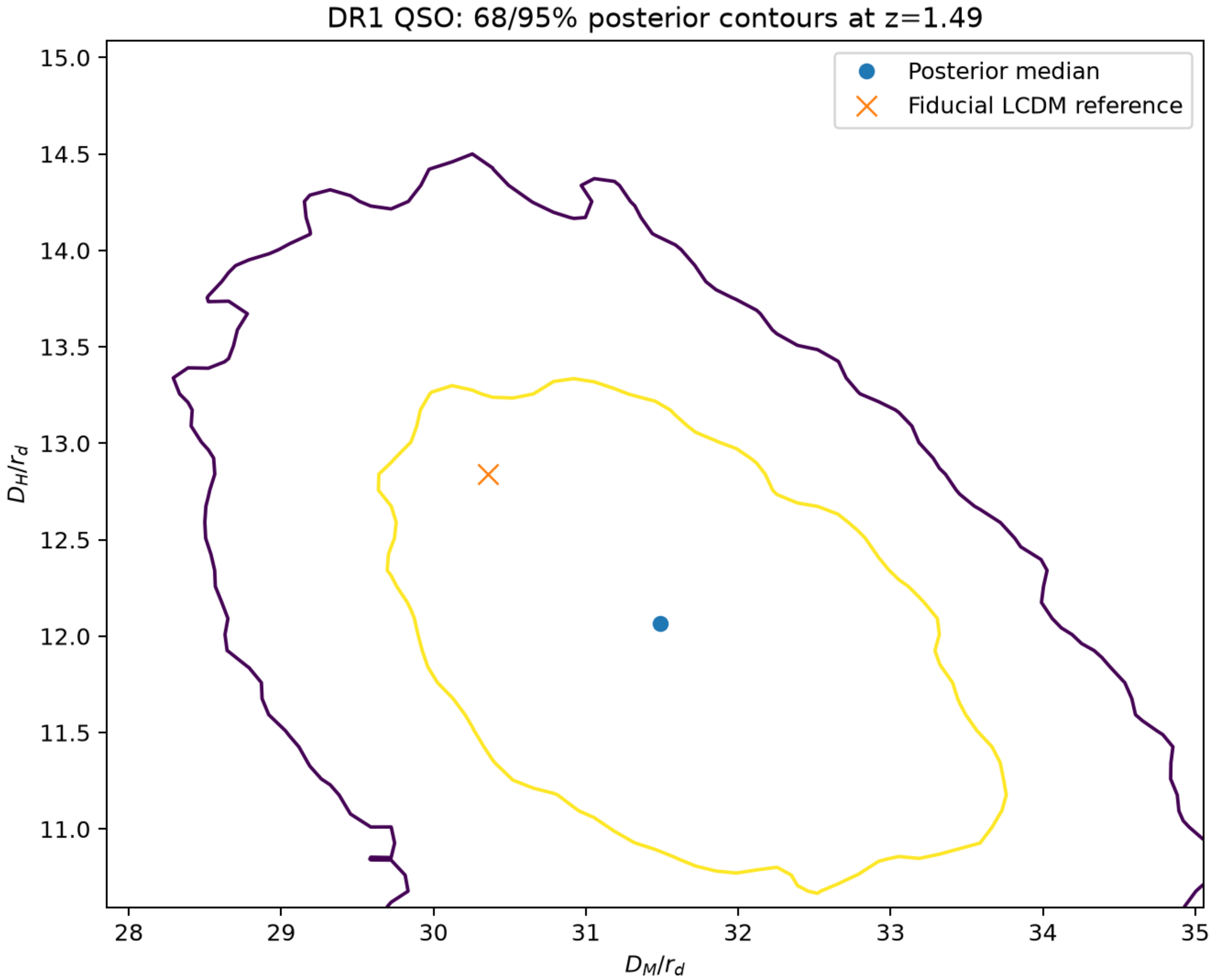
Figure 25 | quadratic d2DH dz2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_d2DH_dz2.png

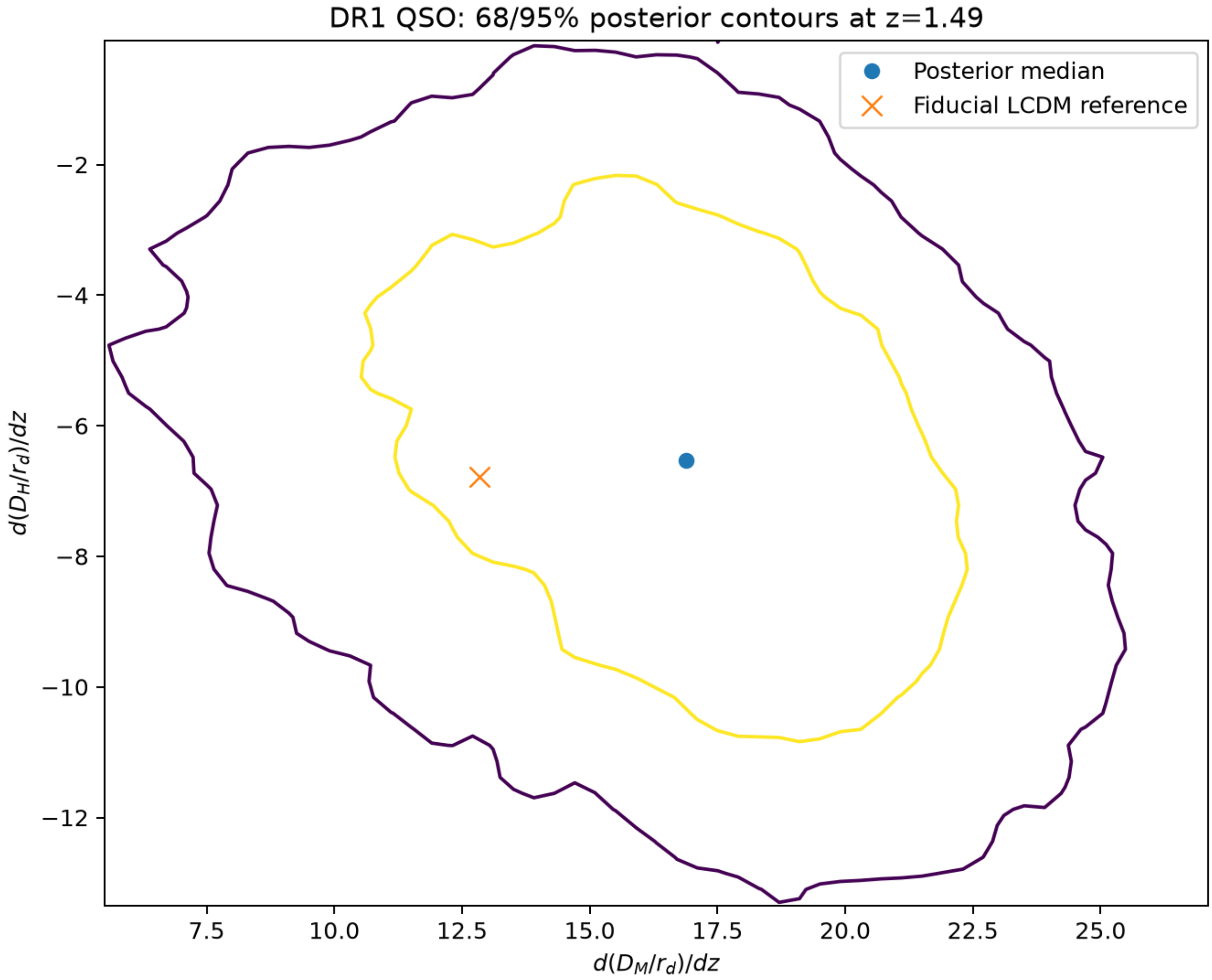
Figure 26 | quadratic joint anchors



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

quadratic_joint_anchors.png

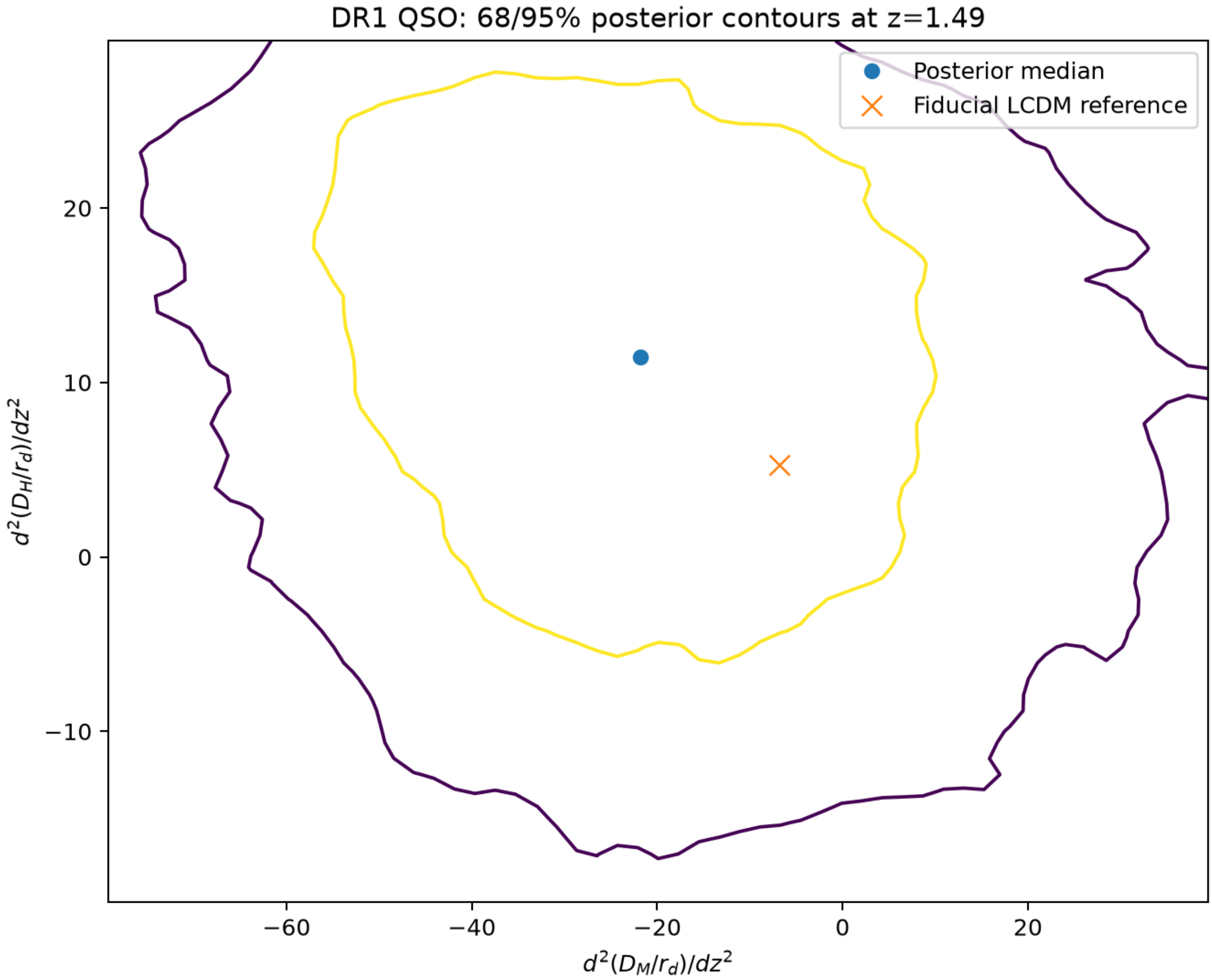
Figure 27 | quadratic joint first derivatives



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

quadratic_joint_first_derivatives.png

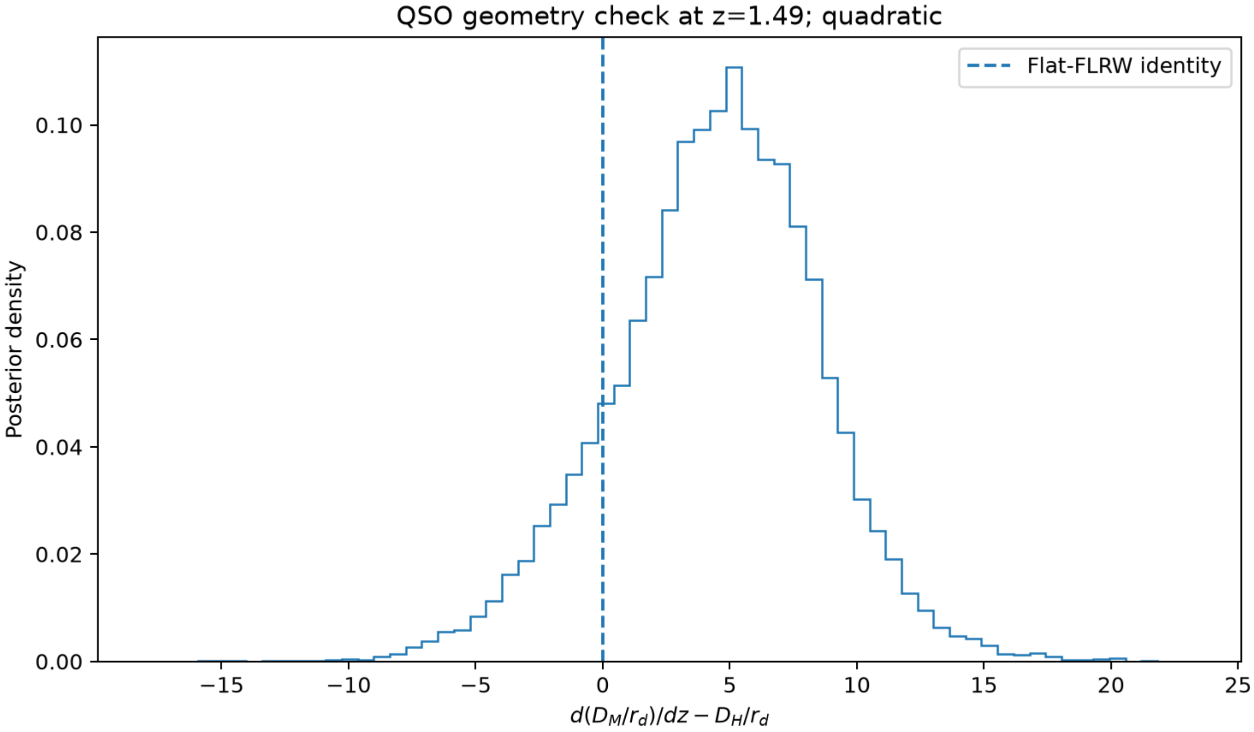
Figure 28 | quadratic joint second derivatives



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

quadratic_joint_second_derivatives.png

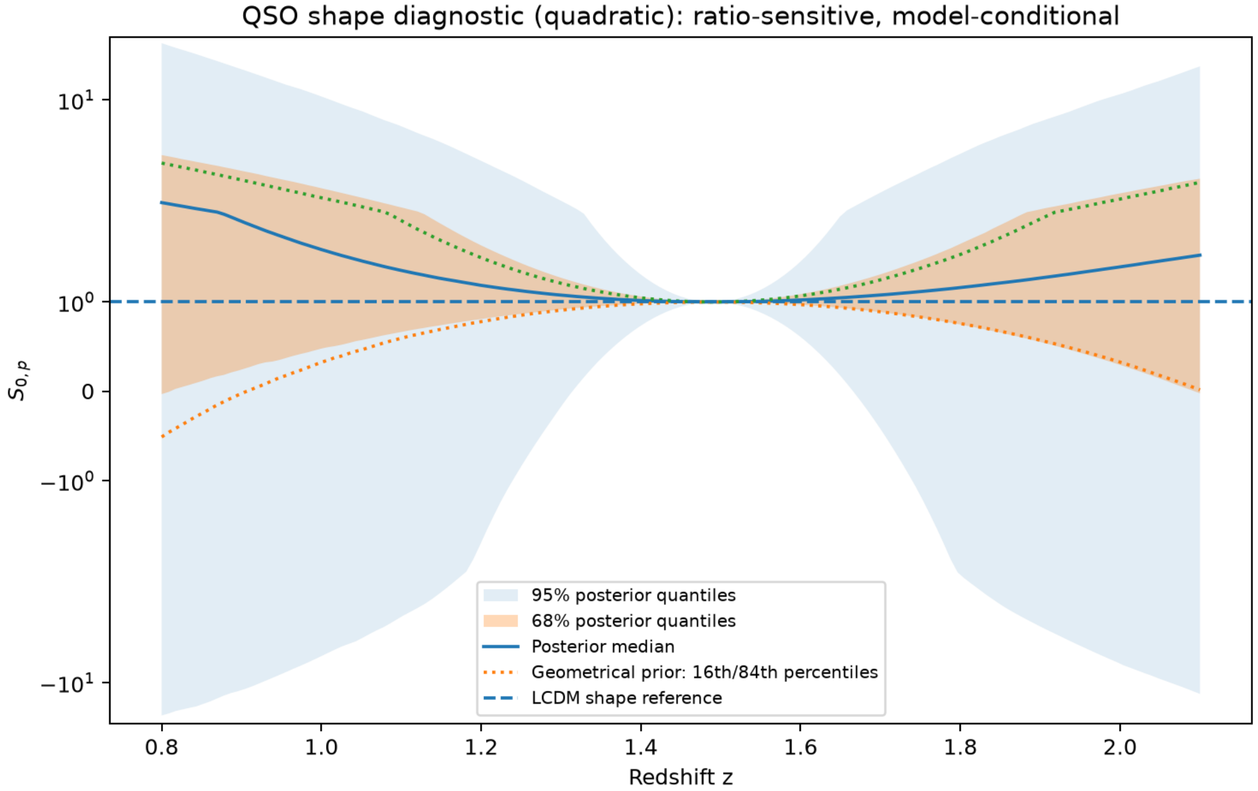
Figure 29 | quadratic flat geometry



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

quadratic_flat_geometry.png

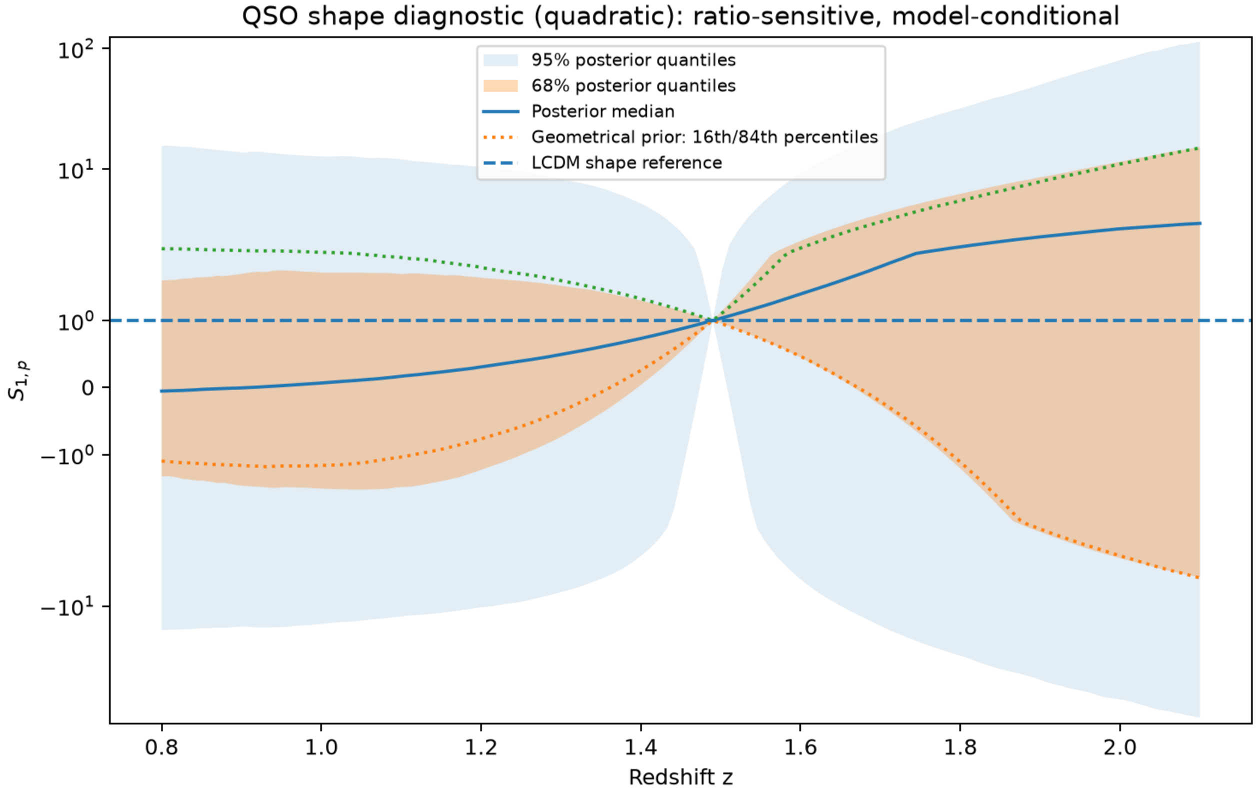
Figure 30 | quadratic S0



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

quadratic_S0.png

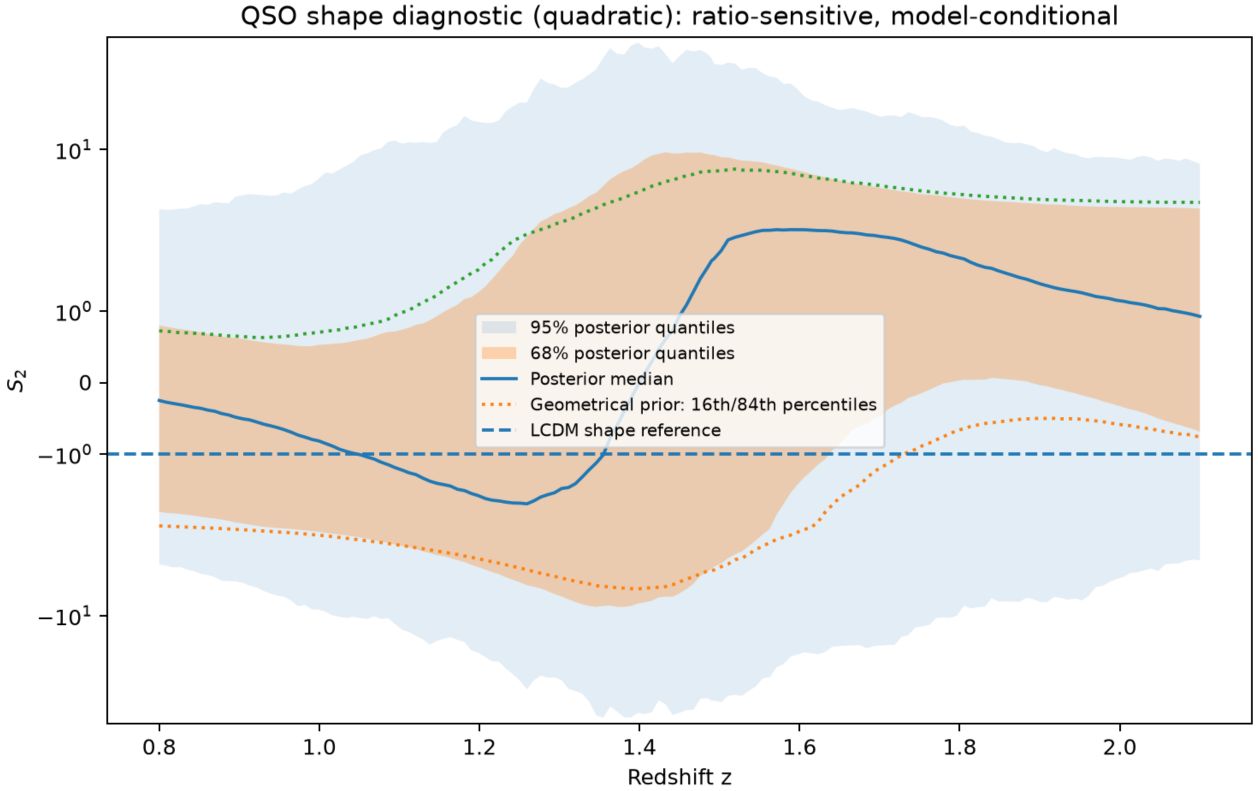
Figure 31 | quadratic S1



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

quadratic_S1.png

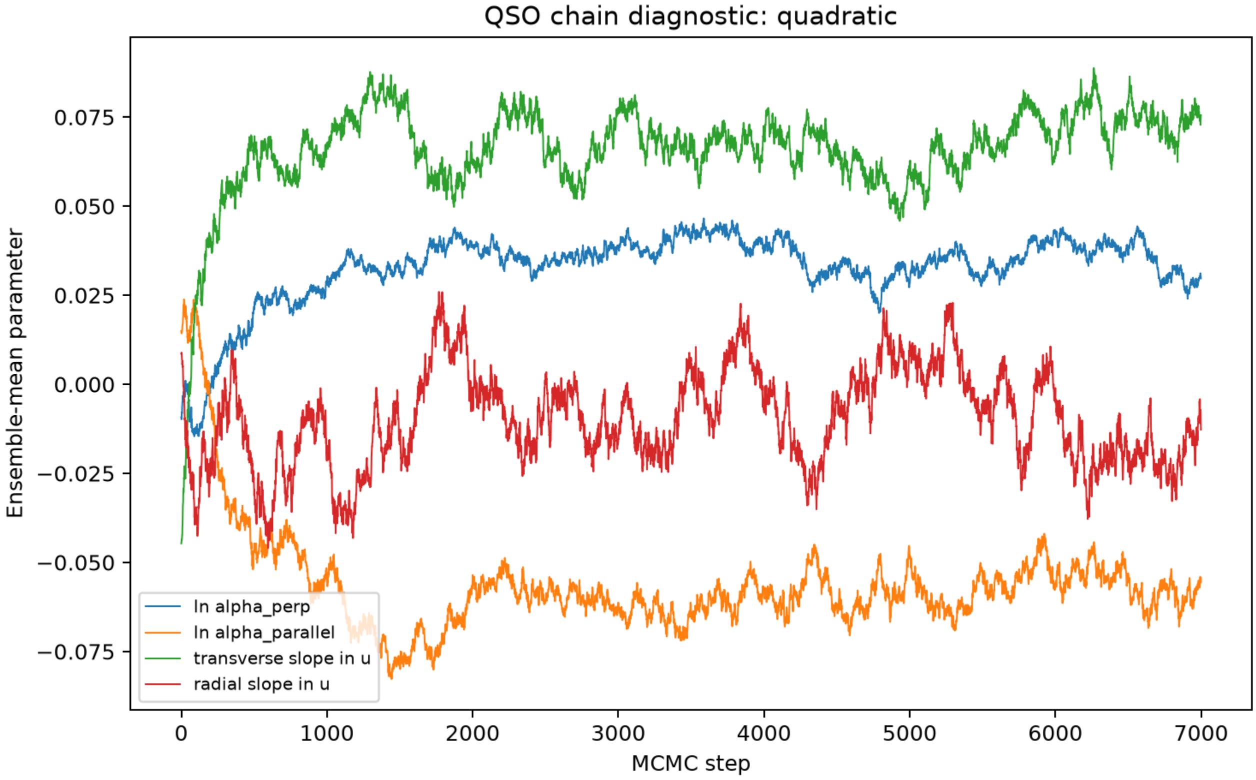
Figure 32 | quadratic S2



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

quadratic_S2.png

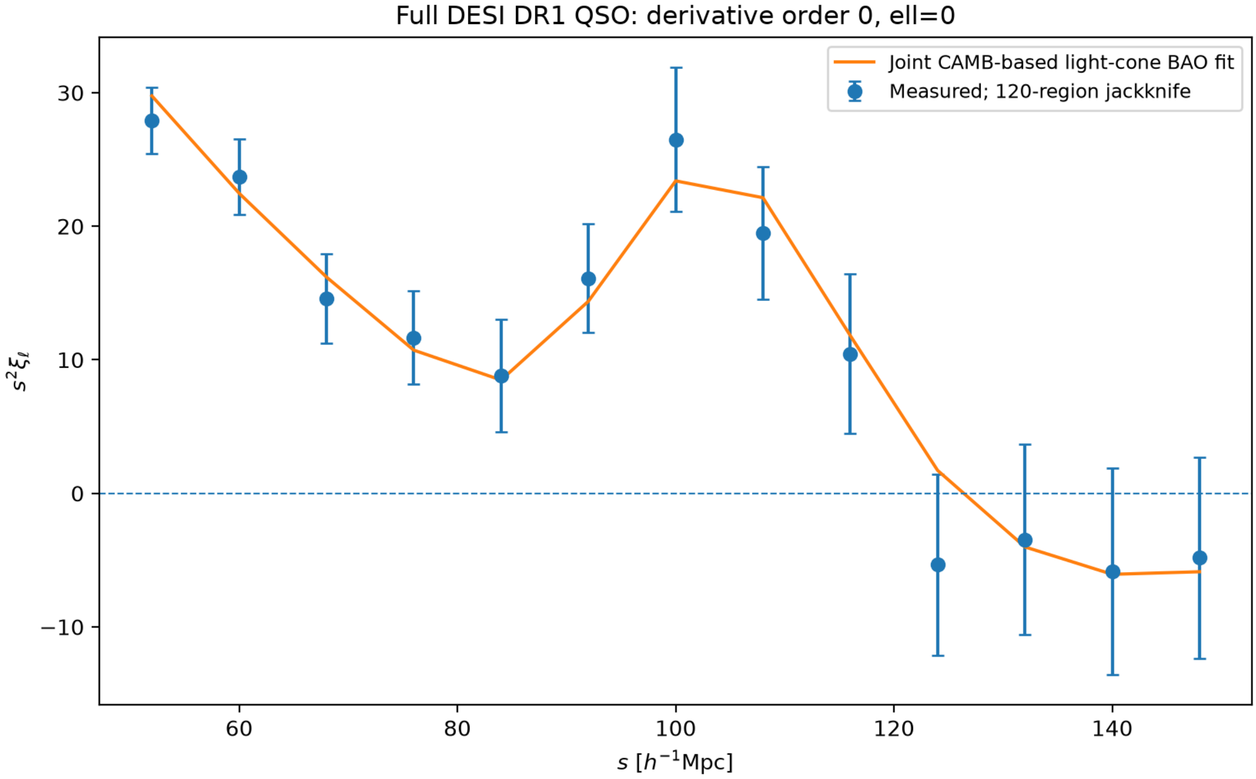
Figure 33 | quadratic chain trace



Ensemble-mean chain traces. These supplement, rather than replace, the saved autocorrelation, independent-ensemble and split-chain diagnostics.

quadratic_chain_trace.png

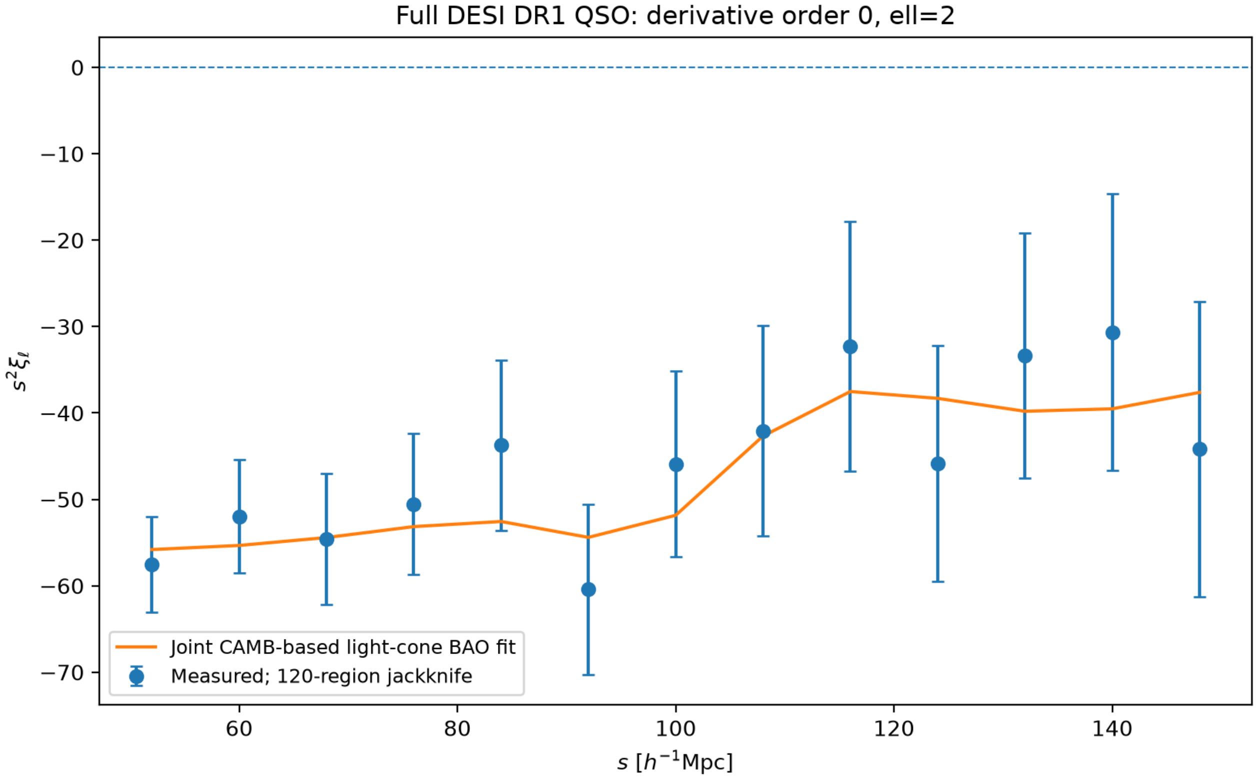
Figure 34 | linear fit order0 ell0



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_fit_order0_ell0.png

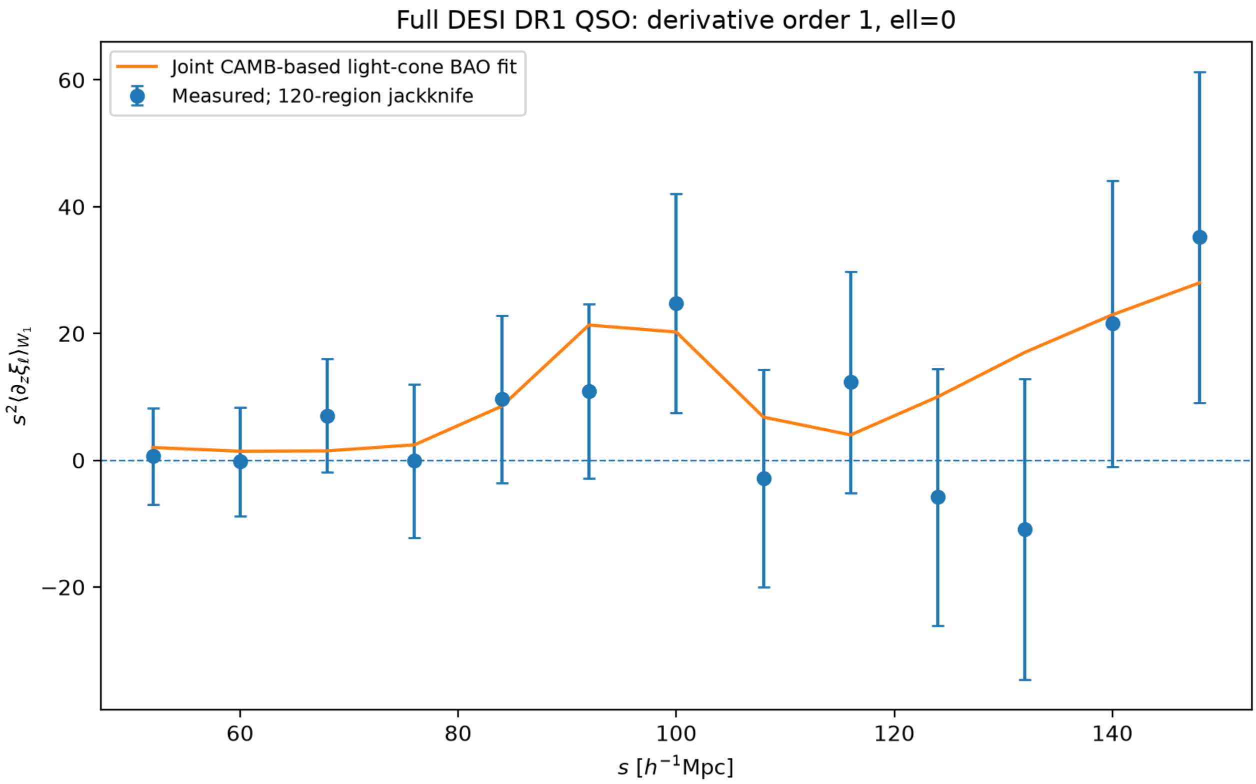
Figure 35 | linear fit order0 ell2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_fit_order0_ell2.png

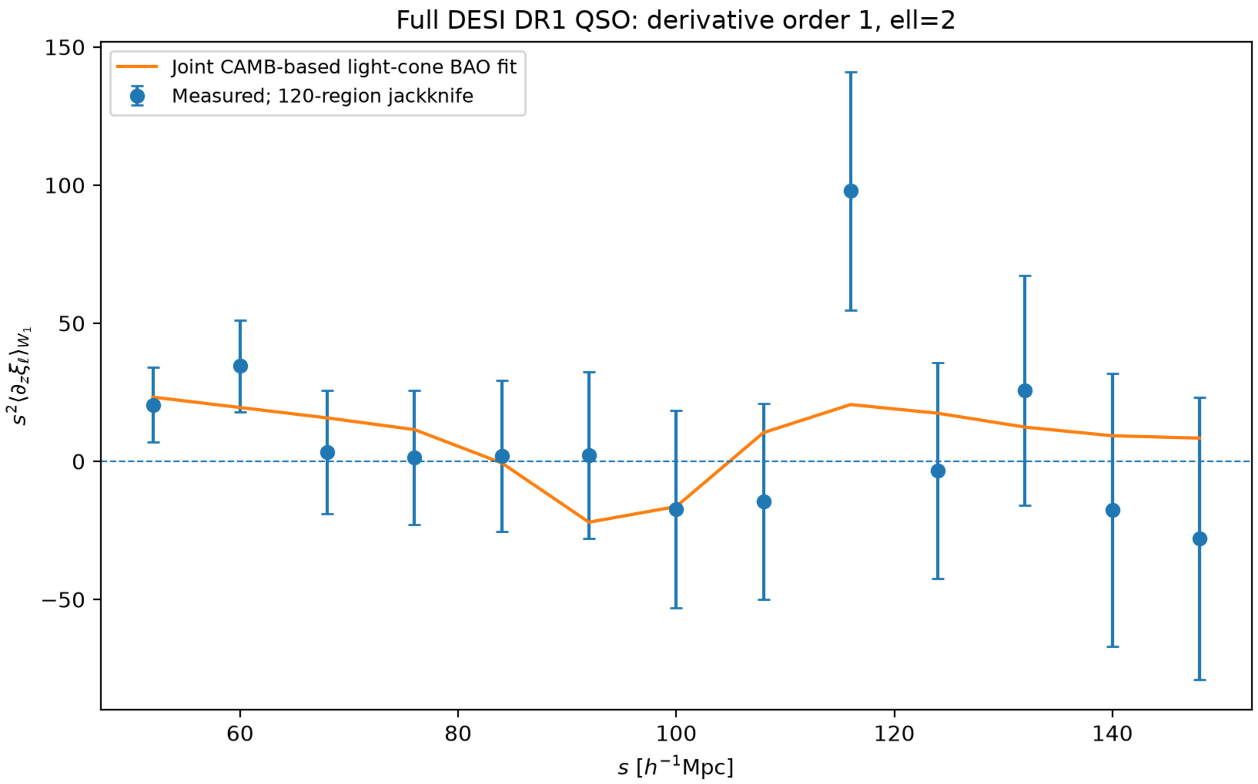
Figure 36 | linear fit order1 ell0



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_fit_order1_ell0.png

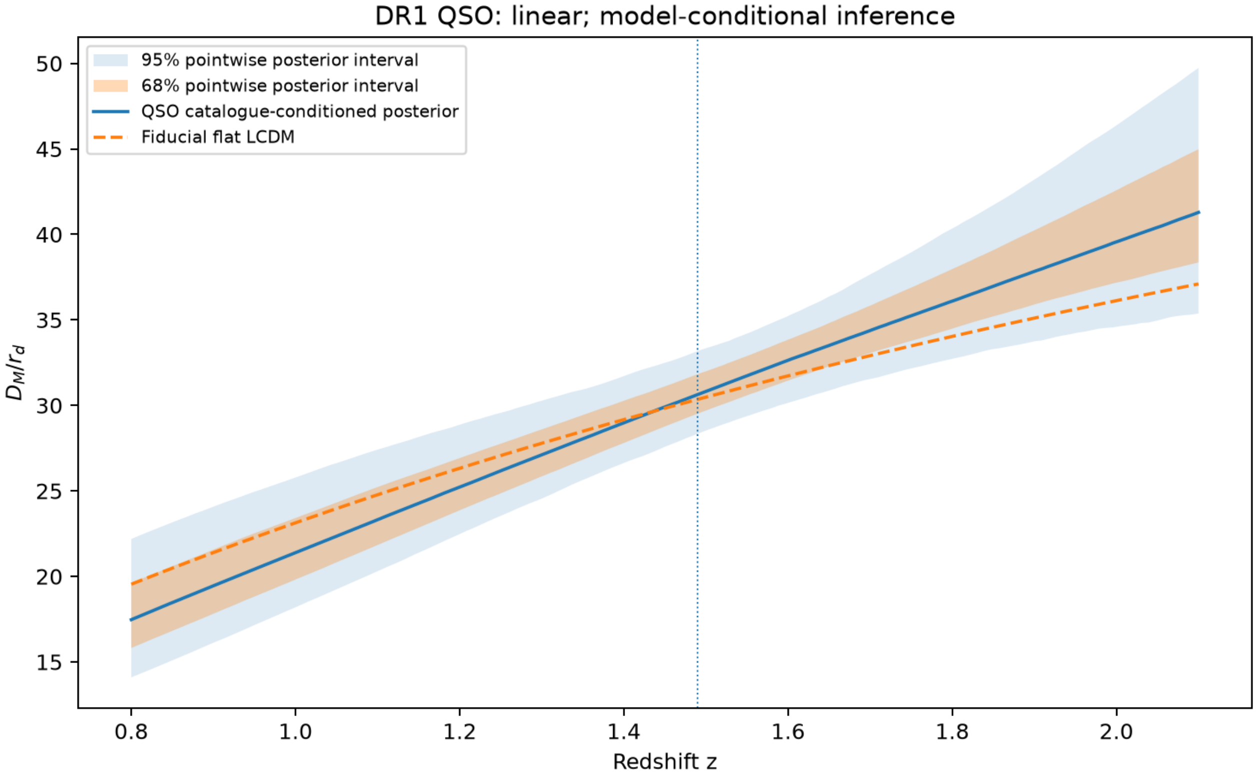
Figure 37 | linear fit order1 ell2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_fit_order1_ell2.png

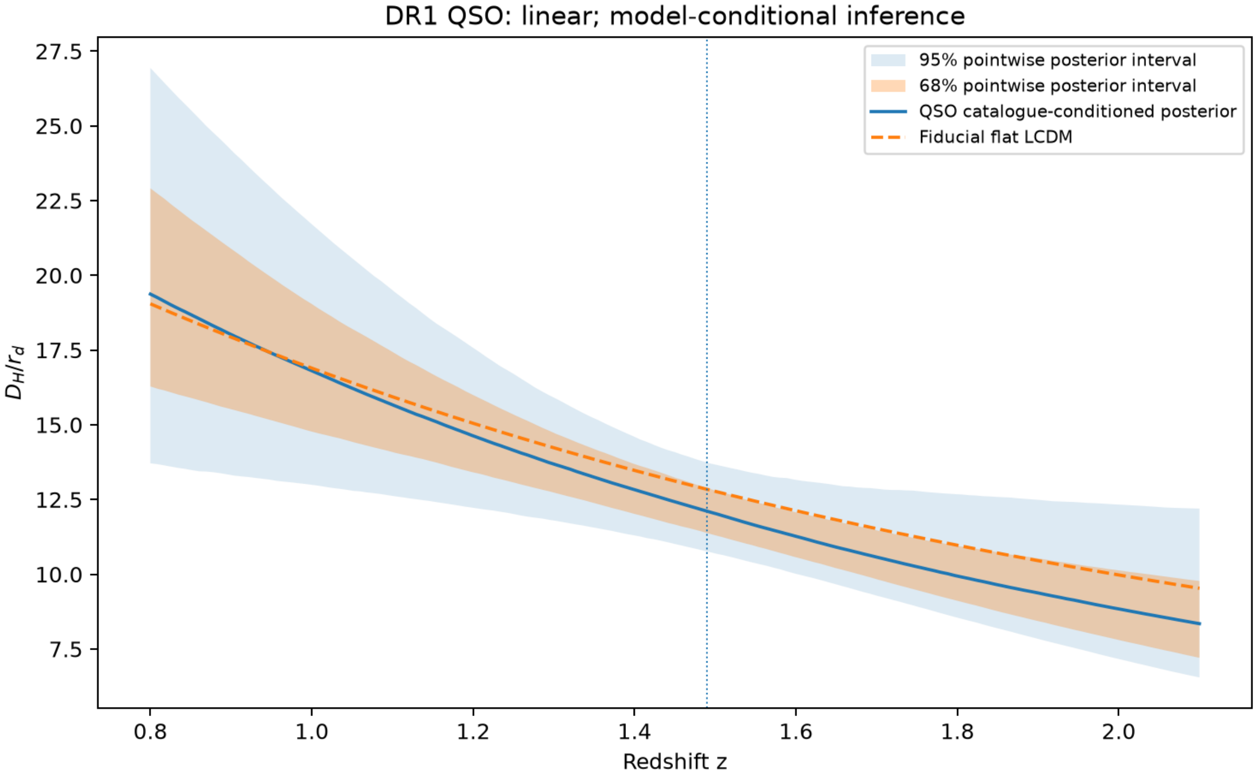
Figure 38 | linear DM



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_DM.png

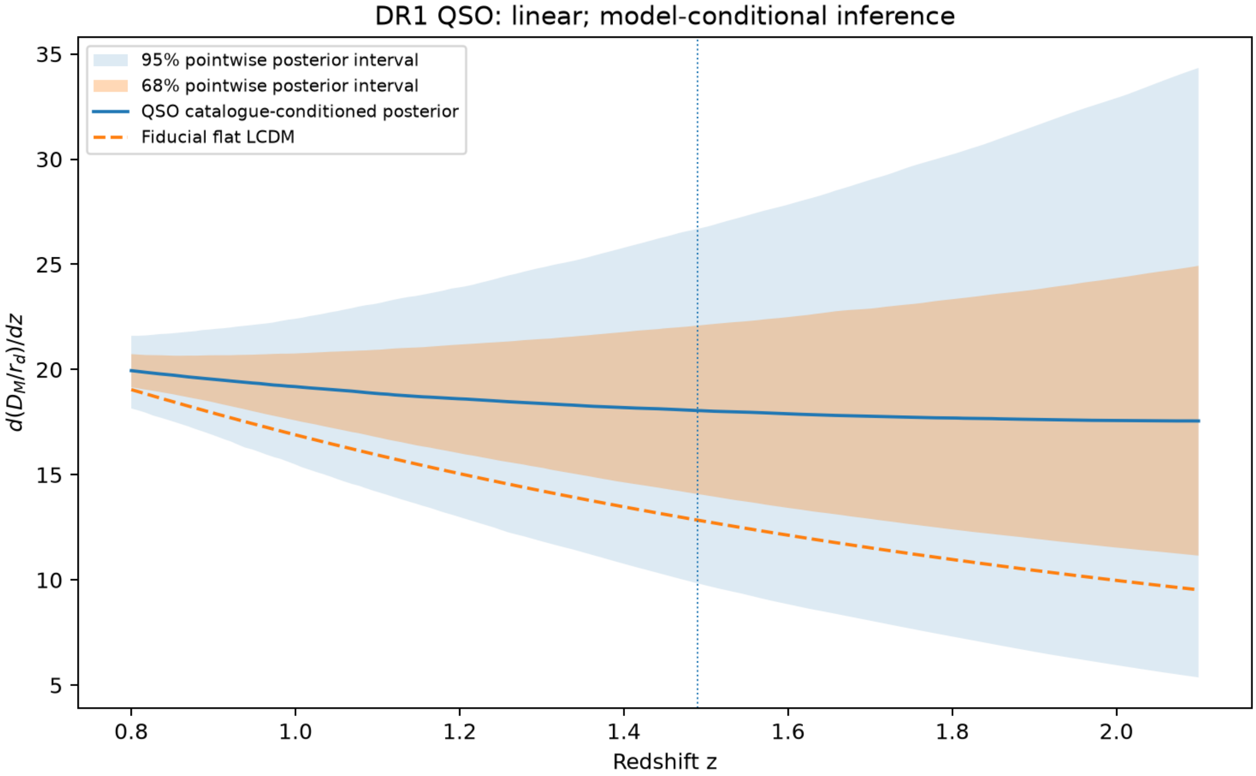
Figure 39 | linear DH



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_DH.png

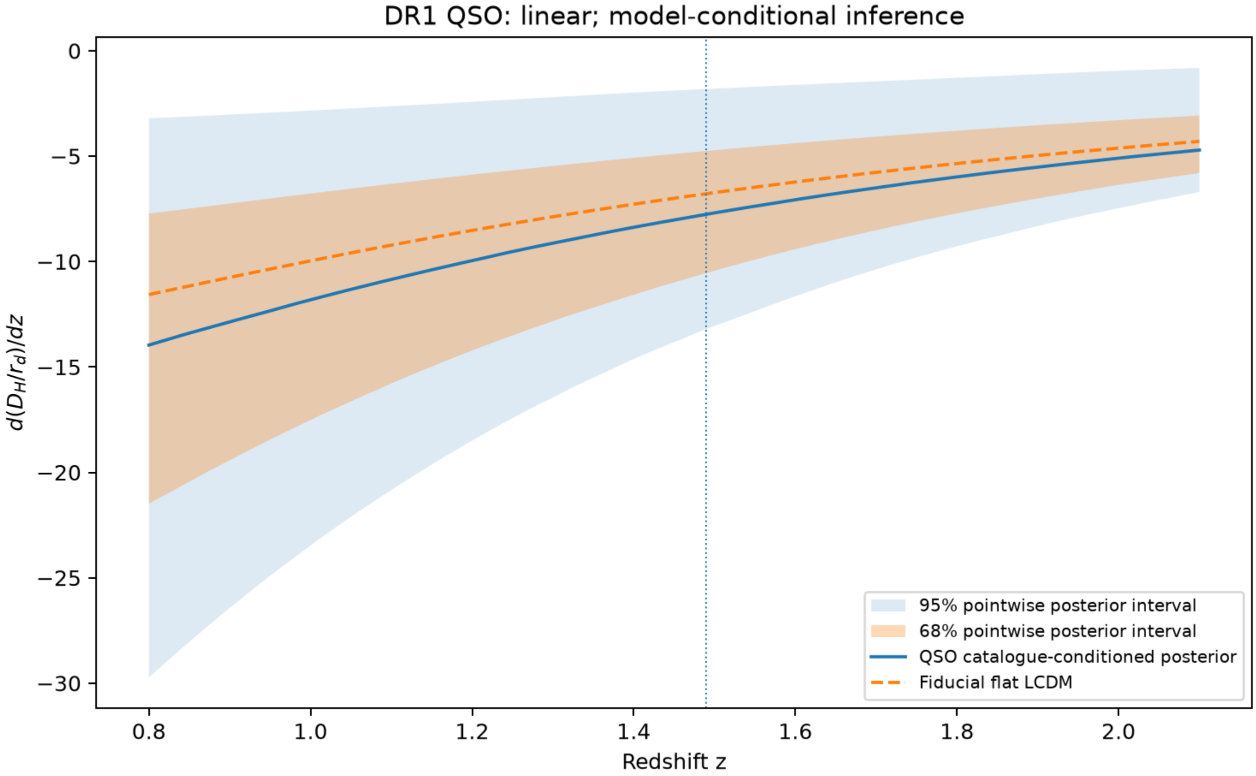
Figure 40 | linear dDM dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_dDM_dz.png

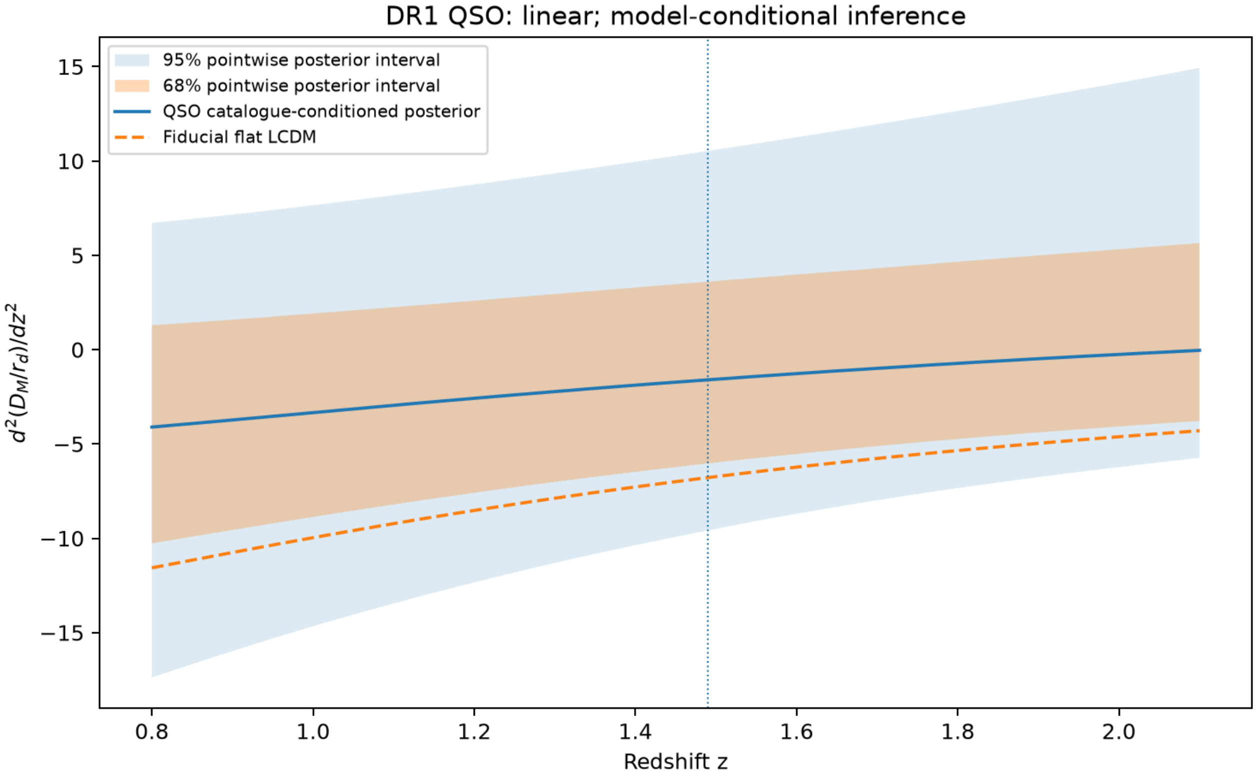
Figure 41 | linear dDH dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_dDH_dz.png

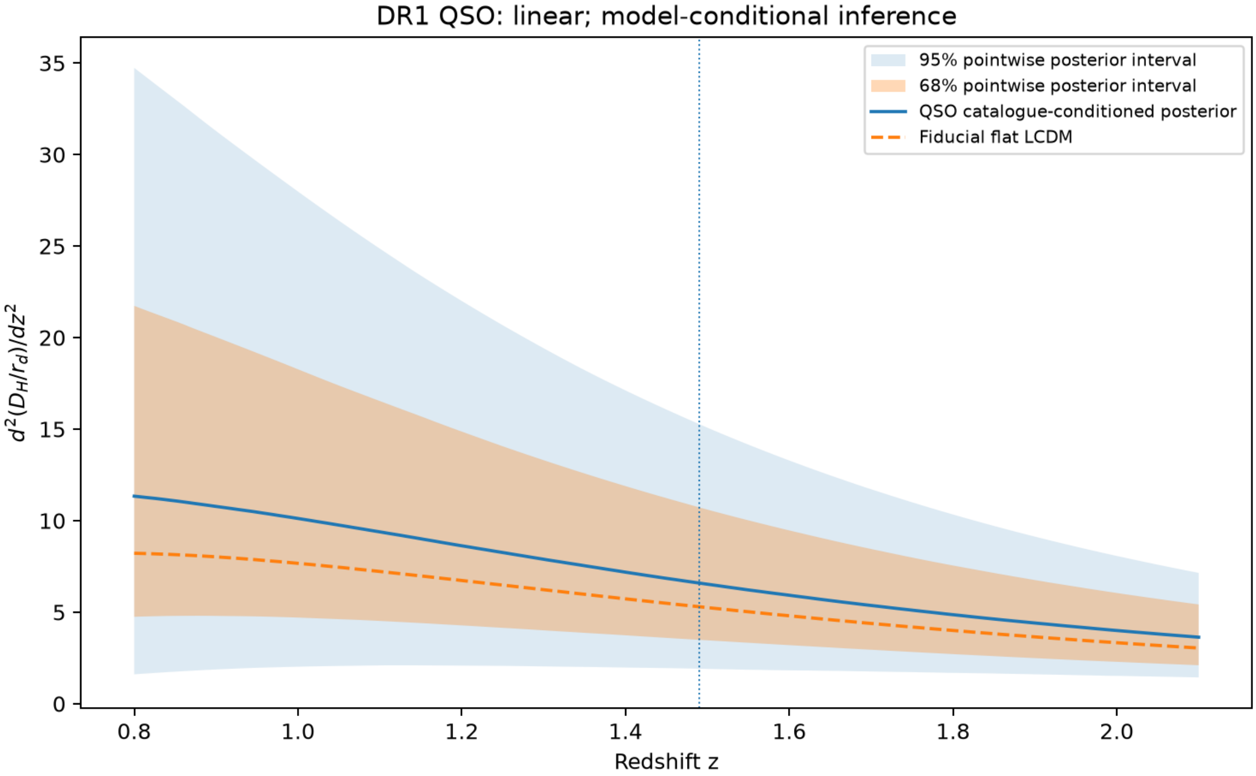
Figure 42 | linear d2DM dz2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_d2DM_dz2.png

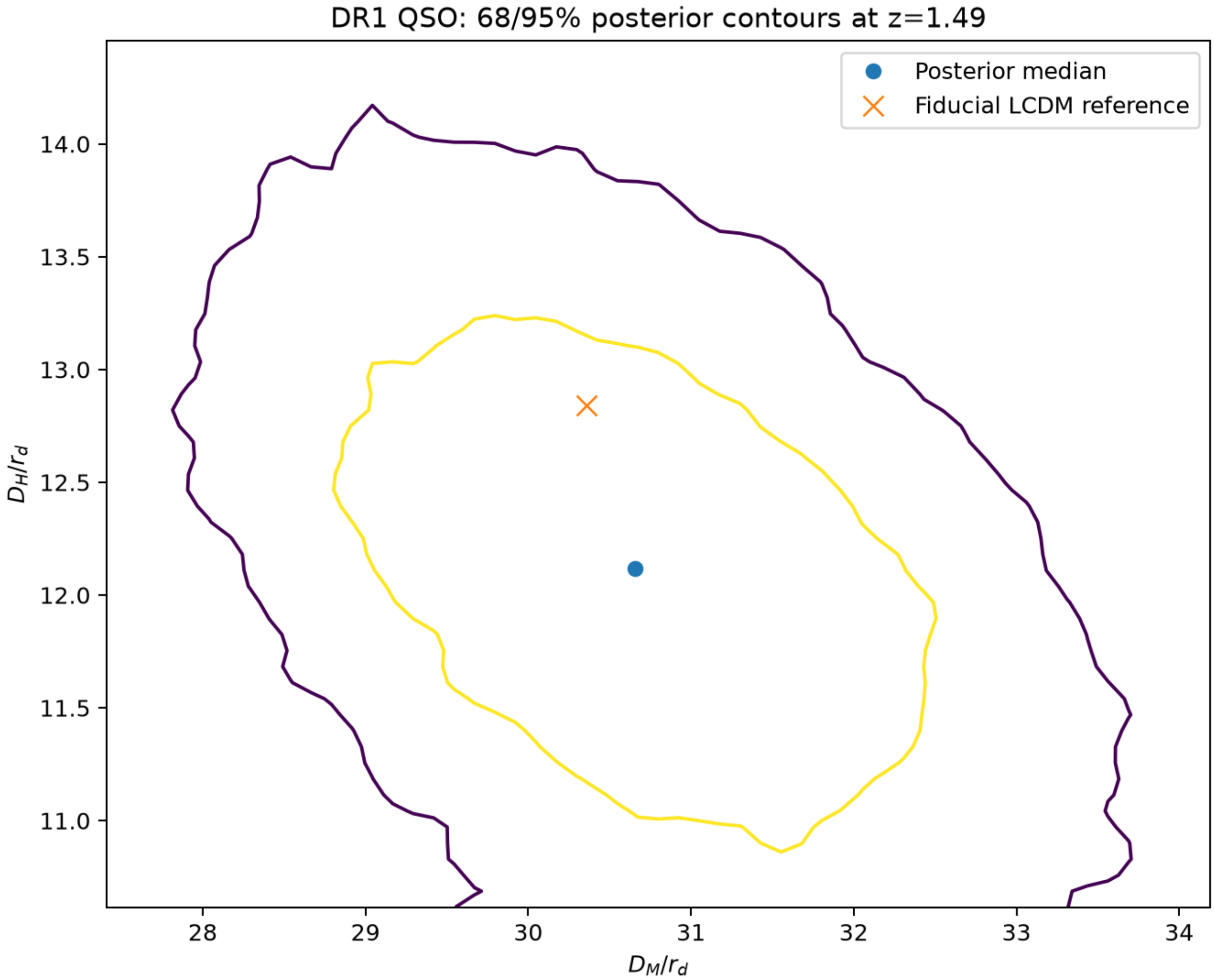
Figure 43 | linear d2DH dz2



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_d2DH_dz2.png

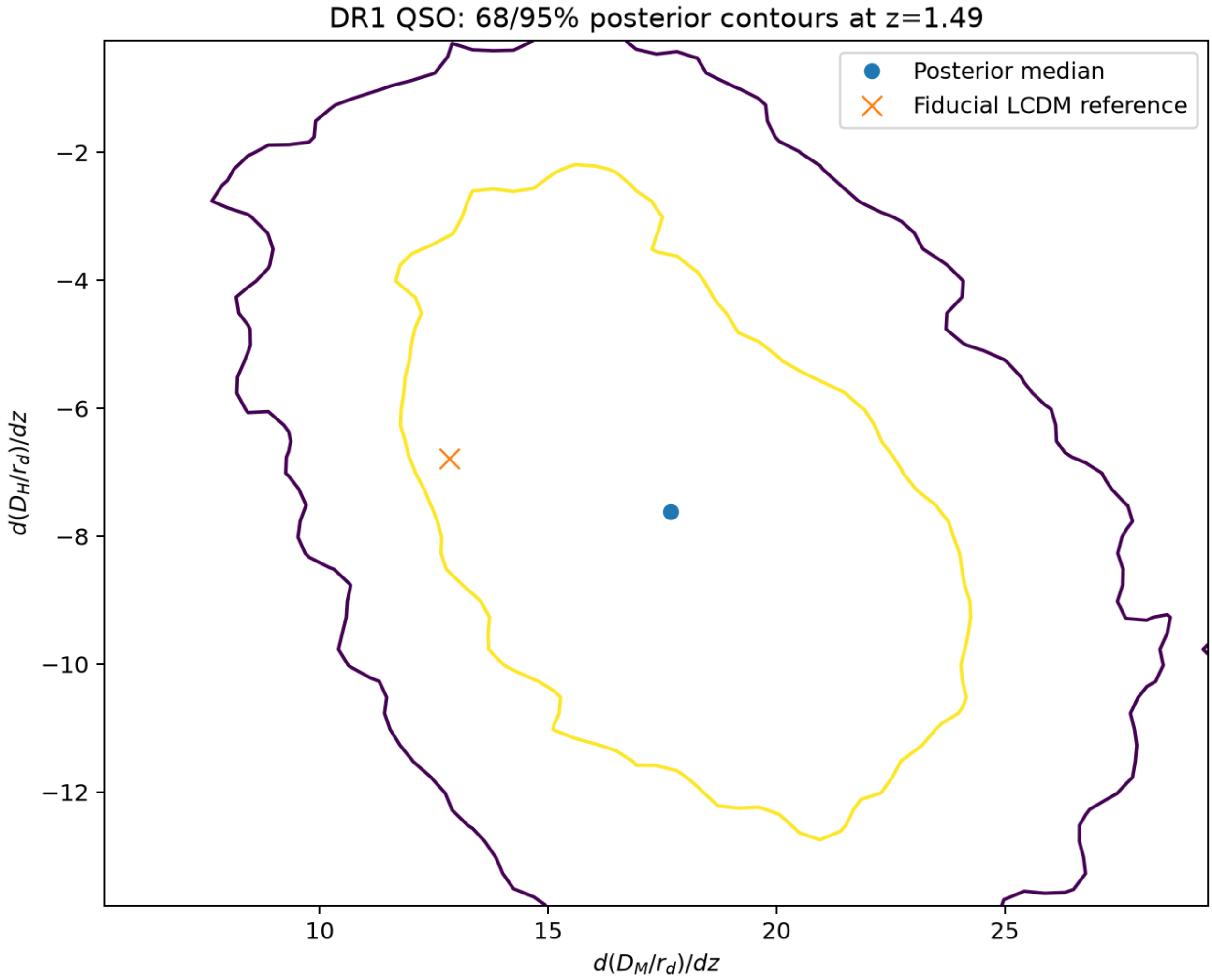
Figure 44 | linear joint anchors



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

linear_joint_anchors.png

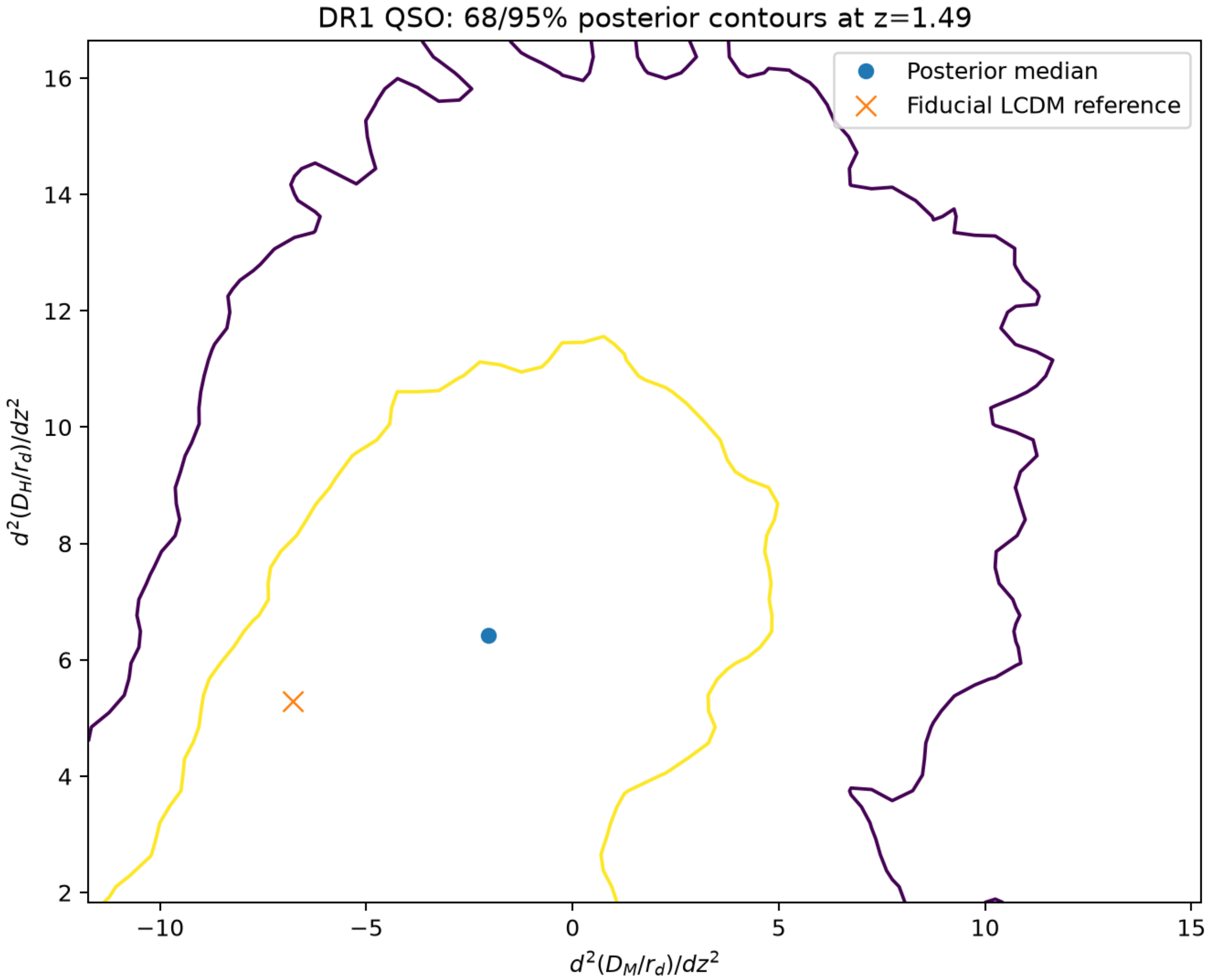
Figure 45 | linear joint first derivatives



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

linear_joint_first_derivatives.png

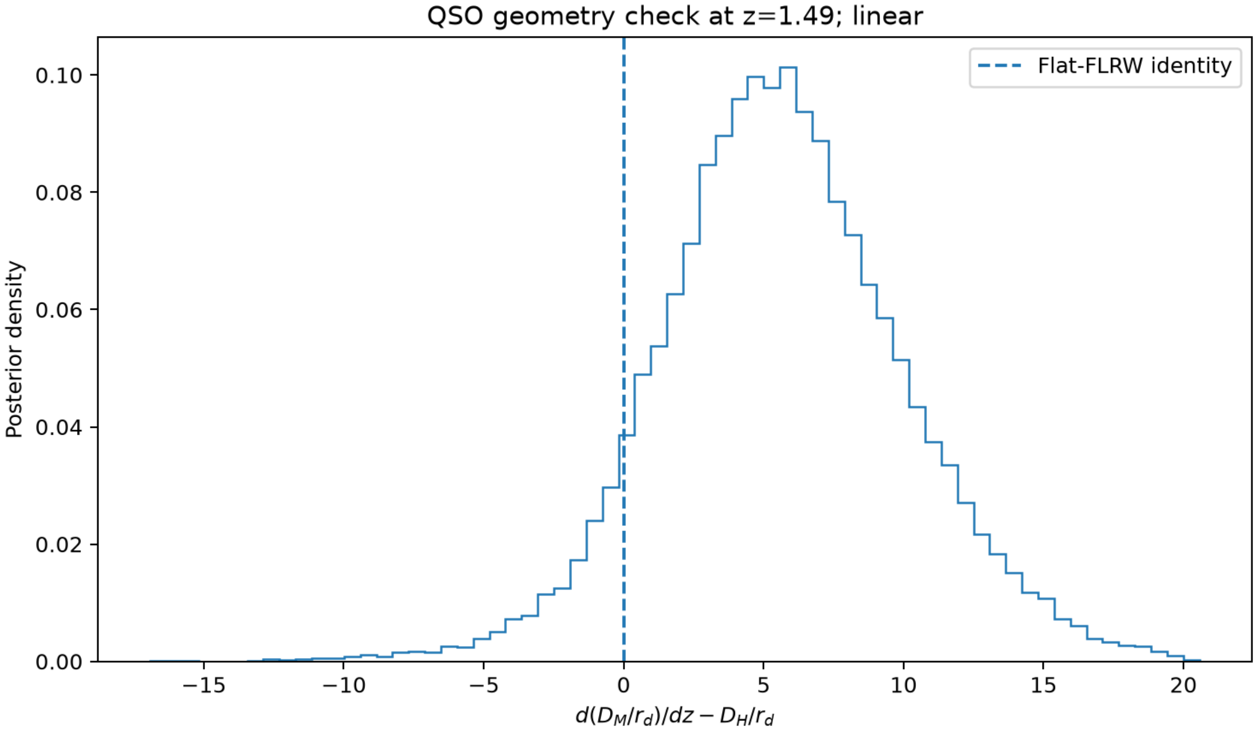
Figure 46 | linear joint second derivatives



Approximate 68% and 95% marginal highest-density contours from smoothed posterior histograms. They include the stated covariance shrinkage and nuisance priors, not independent-mock uncertainty calibration.

linear_joint_second_derivatives.png

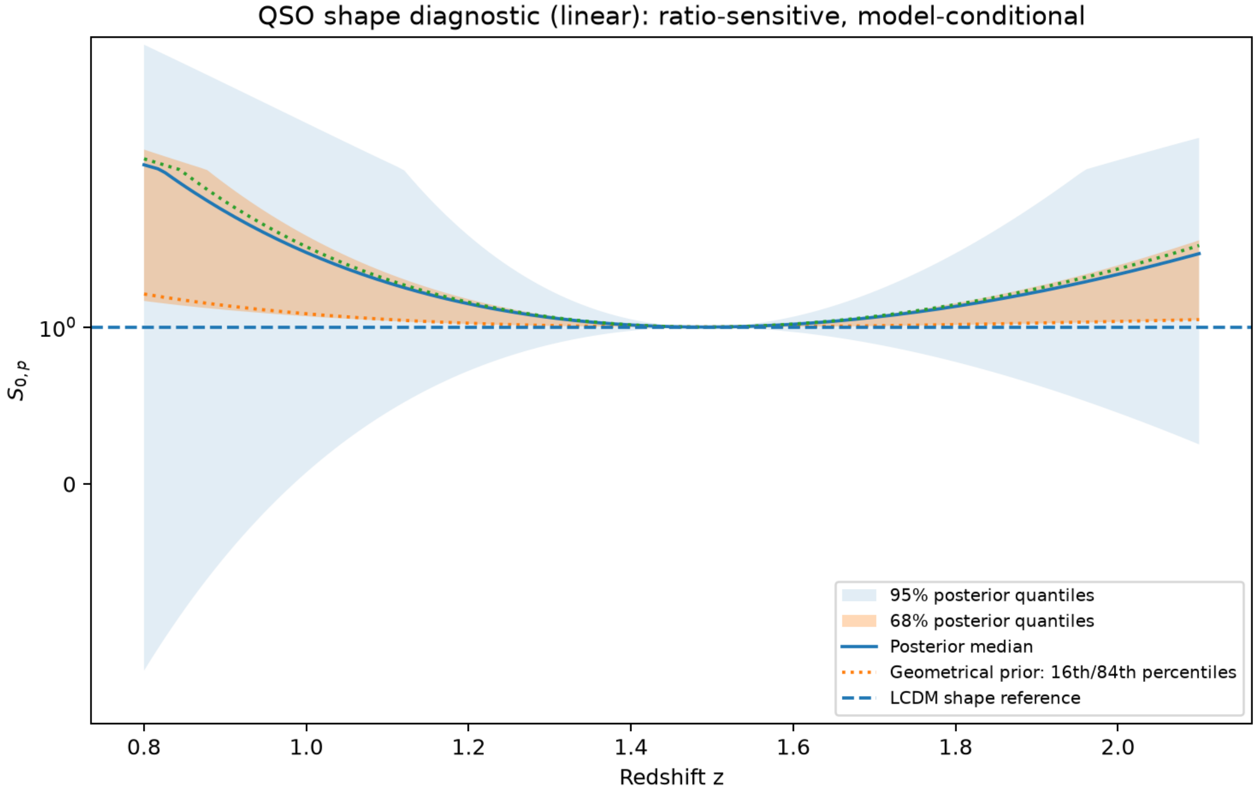
Figure 47 | linear flat geometry



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

linear_flat_geometry.png

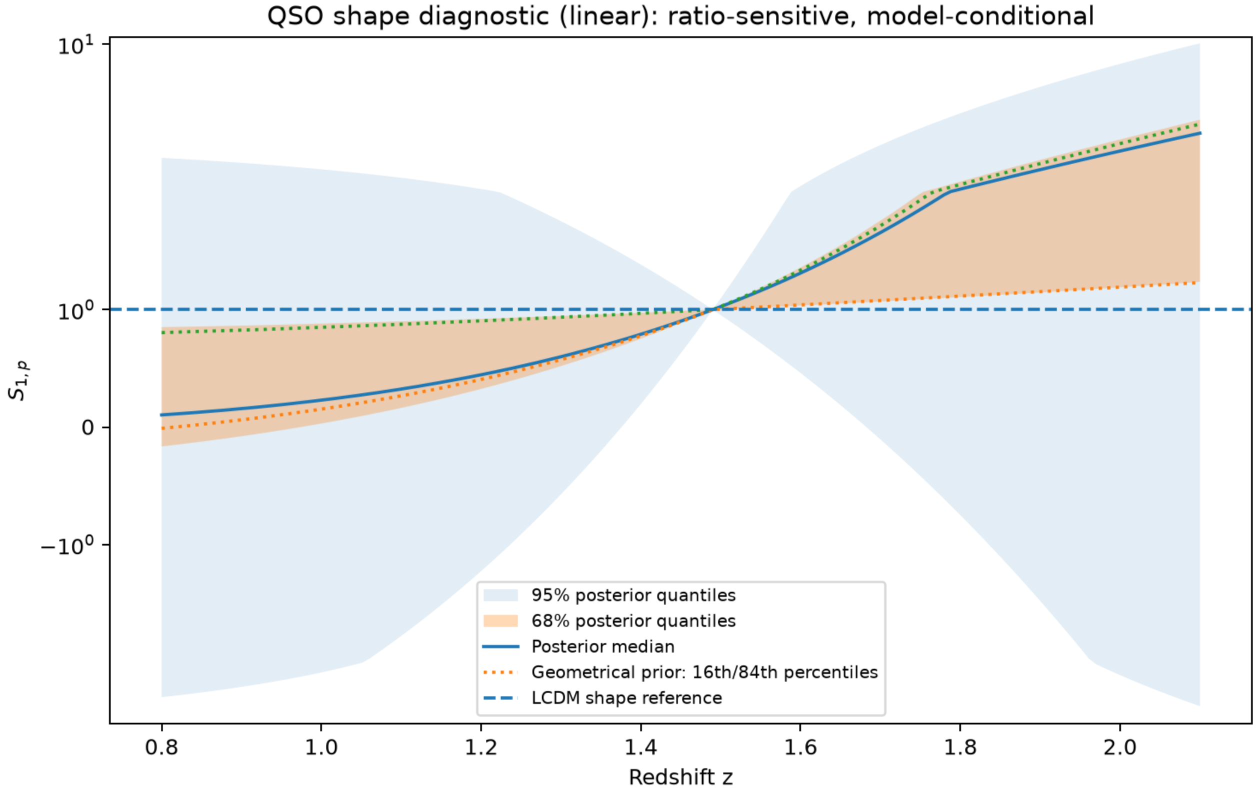
Figure 48 | linear S0



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

linear_S0.png

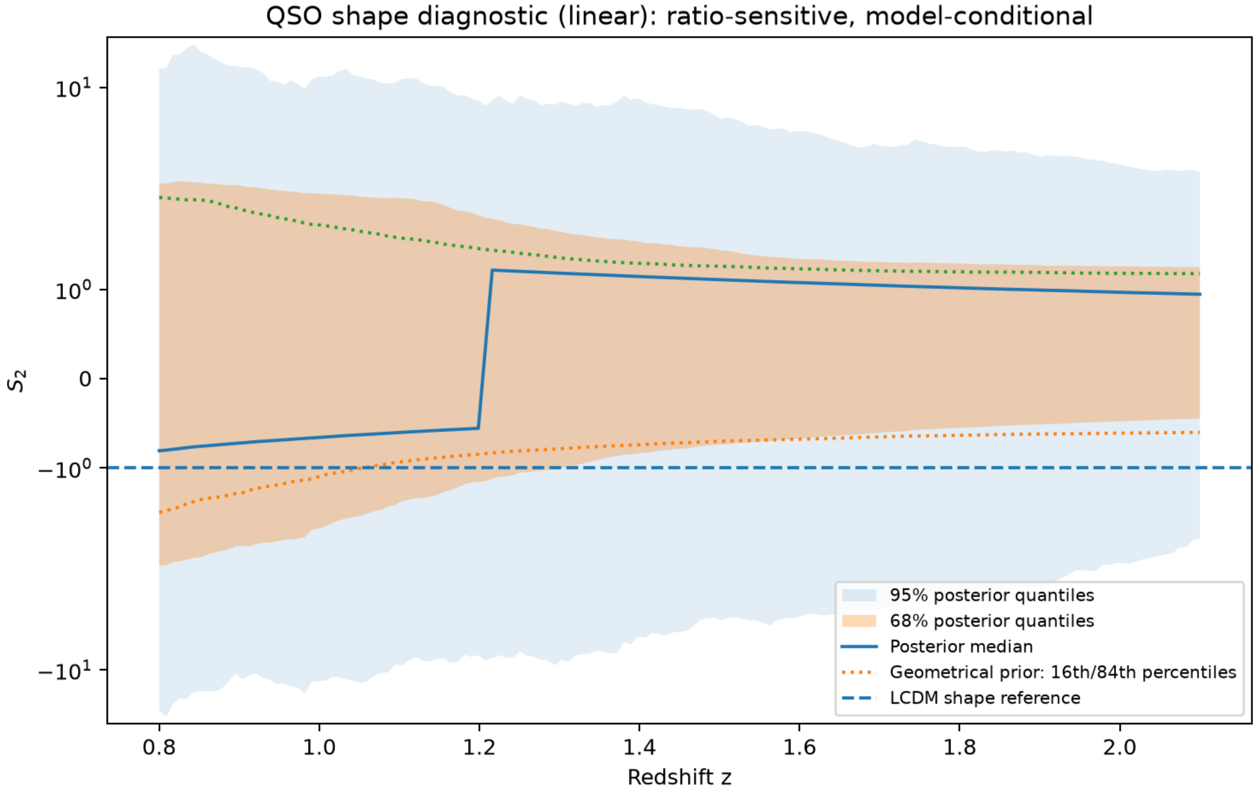
Figure 49 | linear S1



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

linear_S1.png

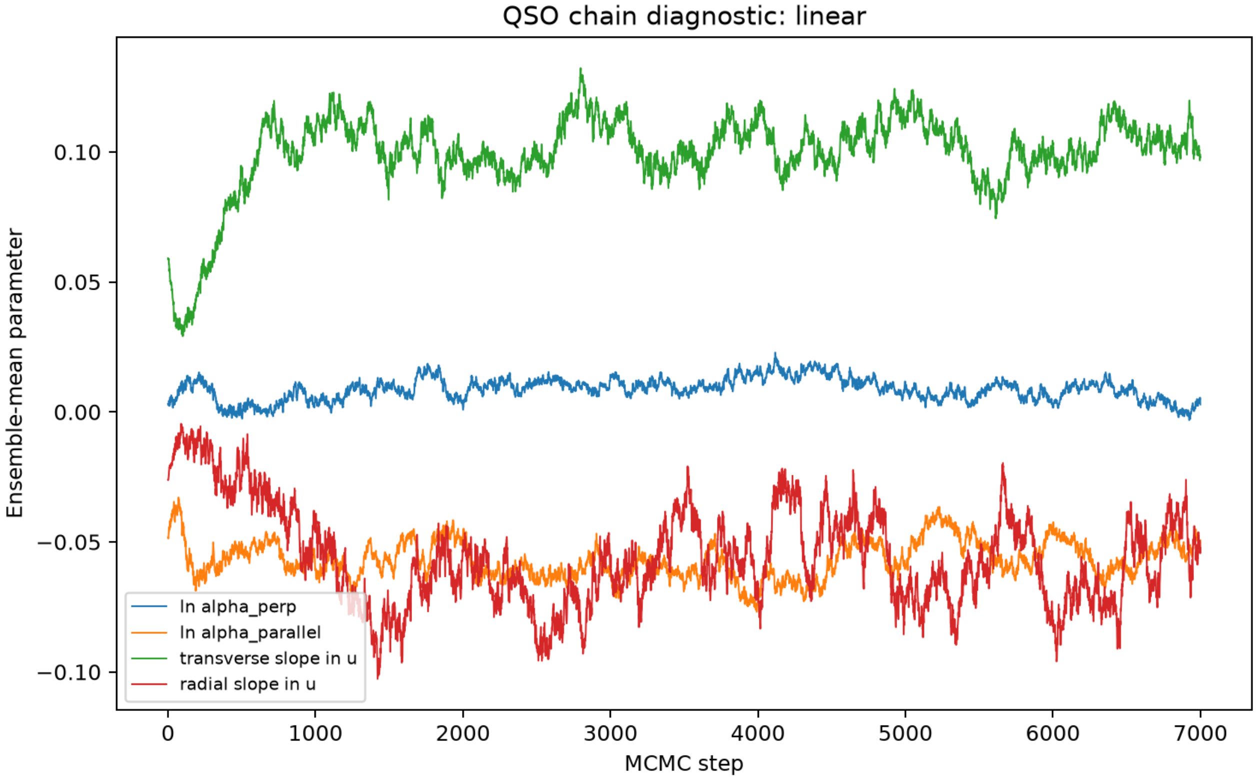
Figure 50 | linear S2



Model-conditional ratio diagnostic. Solid/shaded posterior quantiles are compared with the geometrical prior. Signed-log scaling displays heavy tails. F_x zero crossings are retained. Pivot normalization, not data, fixes S0p and S1p to one at z=1.49.

linear_S2.png

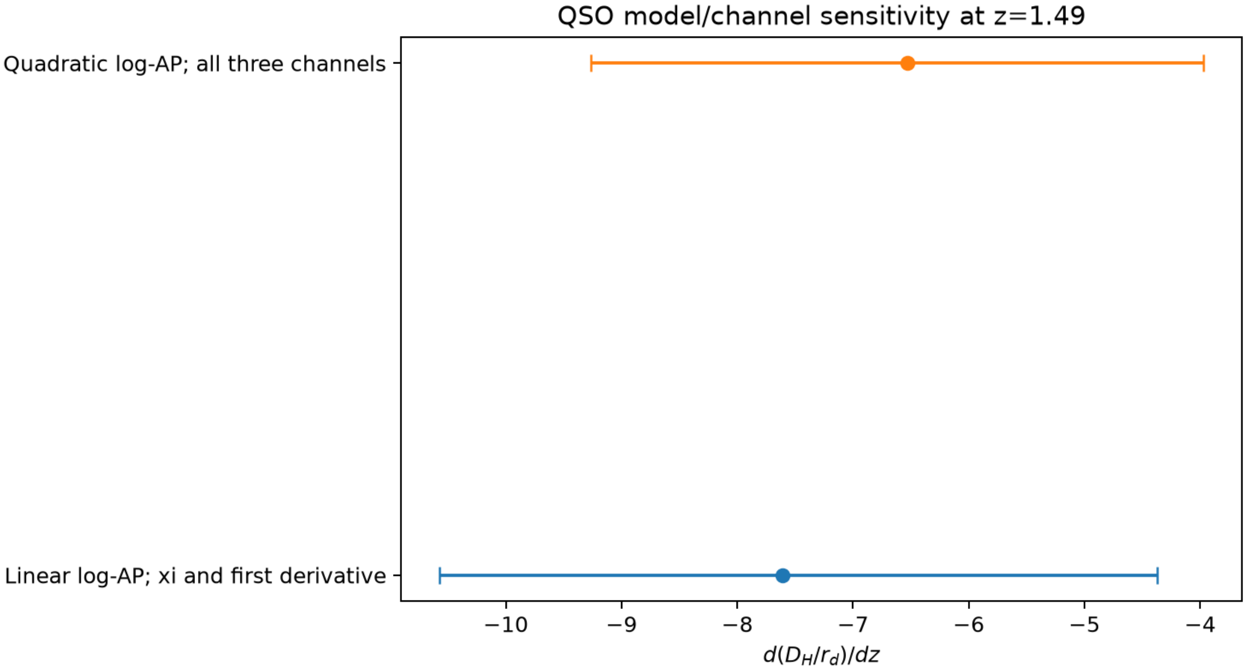
Figure 51 | linear chain trace



Ensemble-mean chain traces. These supplement, rather than replace, the saved autocorrelation, independent-ensemble and split-chain diagnostics.

linear_chain_trace.png

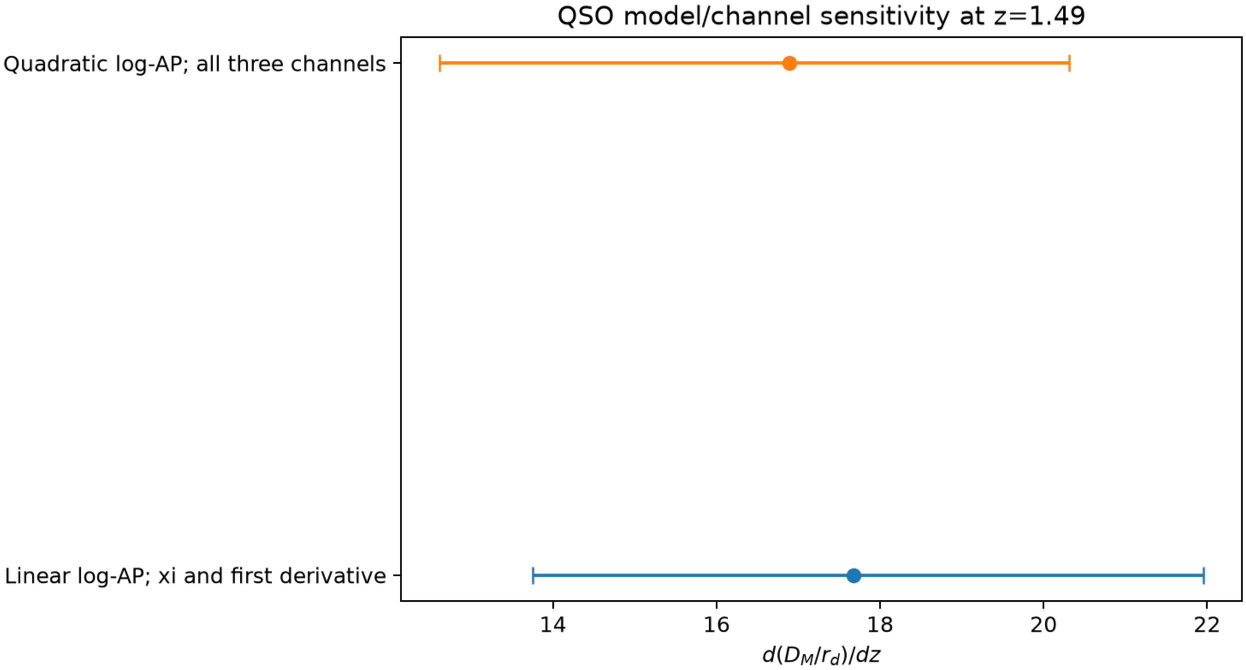
Figure 52 | model comparison dDH over rd dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

model_comparison_dDH_over_rd_dz.png

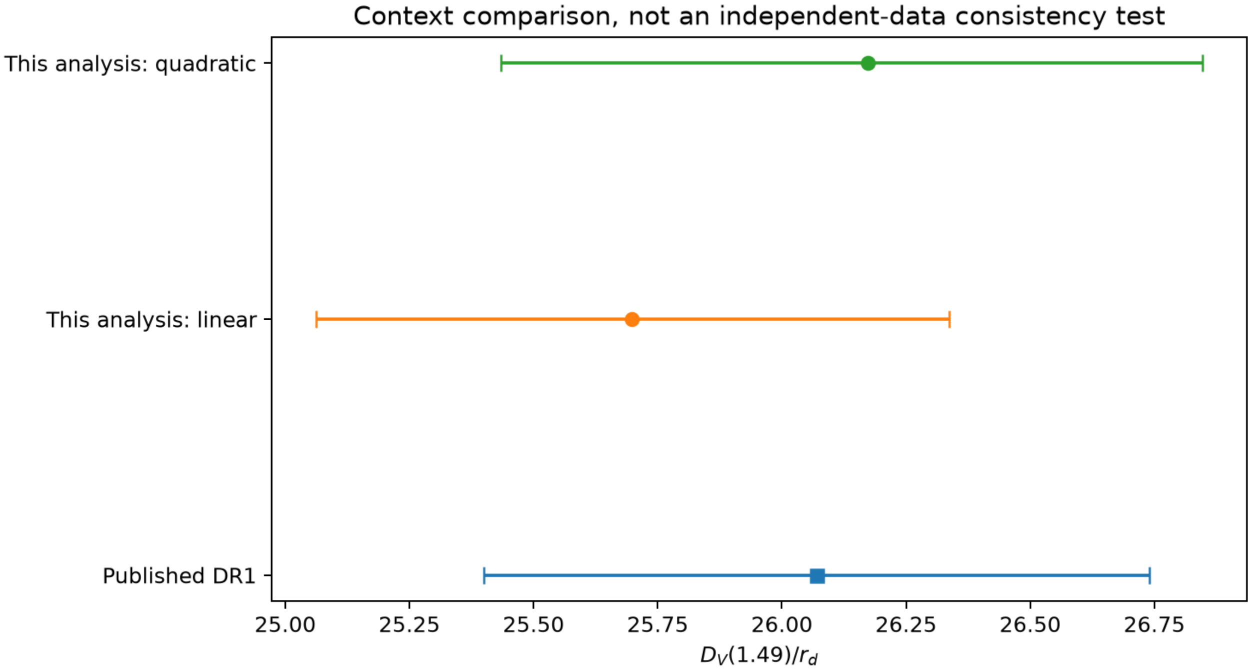
Figure 53 | model comparison dDM over rd dz



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

model_comparison_dDM_over_rd_dz.png

Figure 54 | official DV context



Exploratory QSO catalogue inference. Redshift basis, nuisance model, effective radial smearing and covariance regularization are specified in the method section. Dashed LCDM curves are references, not independent data.

official_DV_context.png